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FACULTY OF ENVIRONMENTAL STUDIES

APPLICATION OF GIS FOR MAPPING AND MEASURING CITIZEN'S ACCESS TO WATER SUPPLY IN RWAMAGANA DISTRICT

A Thesis Submitted to the Faculty of Environmental Studies, in partial fulfillment of the requirements for the award of Master of Science in Environmental and Development Studies, Option of Environmental Information System.

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Kigali, May 2022

DECLARATION

I, NDAGIJIMANA Innocent, hereby declare that this thesis entitled: **"*Application of GIS for mapping and measuring citizen's access to water supply in Rwamagana district*"** is my original work and has not been presented for a degree award in any other University.

NDAGIJIMANA Innocent

Date and signature:…….

CERTIFICATION

This is to certify that this thesis entitled**: "Application of GIS for mapping and measuring citizen's access to water supply in Rwamagana district"** was prepared and is submitted by **Innocent NDAGIJIMANA** under my guidance and approval.

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ACCRONMYS AND ABBREVIATIONS

WASAC:	Water and Sanitation Corporation
PO:	Private Operators
EICV:	Integrated Household Living Conditions Survey
GIS:	Geographic Information System
GPS:	Global Positioning System
WPM:	Water Point Mapping
MININFRA:	Ministry of Infrastructure
WSP:	Water Safety Plans
NISR:	National Institute of Statistics of Rwanda
NST:	National Strategy for Transformation
PWP:	Public Water Point
JMP:	Joint Monitoring Programme
SDGs:	Sustainable Development Goals
UNDP:	The United Nations Development Programme
RADWQ:	Rapid Assessment of Drinking Water Quality
UNESCO:	United Nations Educational, Scientific and Cultural Organization
UNICEF:	United Nations International Children's Emergency Fund
WTP:	Water Treatment Plant
WSS:	Water Supply Systems

ABSTRACT

Water is a mutual shared natural resource fundamental for life. Nevertheless, measuring water access in Rwanda, like in many other developing countries, has been a questionable subject among water sector stakeholders. The overall objective of this study is thus to map and analyze citizens' access to water supply services in Rwamagana district. In order to hit the target of this study, 14 sectors namely Fumbwe, Gahengeri, Rubona, Munyaga, Munyiginya, Musha, Mwulire and Kigabiro, Gishali, Muhazi, Karenge, Nyakariro, Nzige and Muyumbu supplied by rural and urban water supply systems managed by private operators and WASAC respectively were selected as a case study. Data collection was performed in collaboration with private operators (Ayateke star company Ltd and MKM Ubuzima bwiza Ltd) managing rural water supply systems in Rwamagana districts and Representative of WASAC Rwamagana branch who are the main source of information about pipeline characteristics and water infrastructure location, the process of data collection entailed collection of geographical coordinates using GPS device for each water infrastructure such as water source, Public tap, Water Kiosk, Reservoir tank and pipeline characteristics with their attributes data. Data processing was performed using GIS to produce final detailed layouts and excel sheet with detailed information to enhance the validity and reliability of the results. The results revealed that, households where members use water from an improved source (pipe, protected spring) and get it within 500m and 30 minutes round trip are 52.1% while 27.3% are getting water from an improved source (pipe, protected spring), but get it using more than 500m and more that 30 minutes round trip. The remaining 20.6% households use water from an unimproved source (unprotected spring, etc.) and surface water (lakes, rivers, rocks, dams, valleys, ponds, streams, or irrigation canal, etc.). In terms of the affordability constraint, 80 among 150 surveyed households which are not connected to water network responded that they cannot manage connection fees and cannot afford to pay monthly bill. Muyumbu, Nyakaliro, Nzige, Karenge, Muhazi and Gahengeri sectors are the most less served at 29%, 37%, 38%, 41%,44% and 47% respectively while on the other hand Kigabiro and Munyaga sectors found as fully covered at 100% and the most people are connected to their premises.

Keywords: Water Point Mapping and GIS, water access indicators, water supply, Rwamagana district

CHAPTER ONE: INTRODUCTION

1.0 Introduction

The present chapter presented the background and problem statement related to the study. It also presented the formulated research objectives, research questions and the research significance along with definitions of key terms.

1.1 Background of the study

Water is a mutual common resource fundamental to life and in sustaining the environment as well as enhancing the social and economic development of our wellbeing. It is vital for sustainable socioeconomic development as a strategic primary input playing a central role in poverty alleviation through enhancing food security, domestic hygienic, hydropower, industrial development, mining, navigation, and the environment for sustenance of ecosystems (MININFRA, 2016). Availability of adequate water supply, for drinking and other domestic use, of good quality reduces time spent in fetching water, increases health standards by reducing the incidences of debilitating water-borne diseases such as diarrhea and cholera and thus improve the socio-economic wellbeing of users (UNICEF and WHO, 2006). Owing to its significance to human life, the United Nations General Assembly and the Human Rights Council explicitly recognized the right to safe and clean drinking water and sanitation as a human right in July 2010 (The UN Special Rapporteur, 2014b). It contended further that the UN member states have an obligation to move as quickly and effectively as possible towards full realization of the human right to water using the maximum available resources.

However, actors in the water sector, including international donor agencies, development partners, national governments, civil society organizations, and scholars, have disagreed on how to define and measure water access as an intangible term. Because of the different settings and situations among countries and sector-stakeholders, "one-size-fits-all" water access standards are likely to fail. The core concern is rooted in conceptual and methodological disagreements among and among sector players, where many "indicators" have been developed, adopted, and applied as proxies for defining and measuring water access. The indicators adopted and implemented serve purposes beyond measuring water access as they help to plan and implement sector development interventions or strategies to monitor and evaluate, as well as to report on sector-development

progress. Nonetheless, the identified pitfalls in the design, construction, or selection, use and interpretation of some indicators is still a challenge which is associated with variety of costs and risks to sector-stakeholders (Holzapfel, 2014; Wong, 2006).

As a result, the goal of this research is to improve our understanding of not only the development, selection, usage, and implementation of water access indicators, but also their impact on evaluating residents' access to water in rural areas. Throughout this investigation, it should be obvious that the term "improved drinking-water source" refers to "improved water point" 1 as described in the Joint Monitoring Programme (JMP) report, and the researcher will use the term "water point" to signify something similar to JMP's definition (WHO and UNICEF, 2005). The term 'public water point' also refers to water sources that serve the general population.

In January 2018, MNINFRA commissioned the development of a web-based Water, Sanitation and Hygiene (WASH) Management Information System (MIS) which was completed in June 2019. The WASH MIS has been used to collect data on WASH services where so far, the data collection has been completed for 17 districts. In trying to address the methodological challenges in determining water access, the government has adopted and is implementing the Water Point Mapping (WPM) system developed by an international organization WaterAid. According to Jiménez and Pérez-Foguet (2008a) WPM methodology allows the definition of more reliable access indicators, in terms of physical coverage, which are centered on water point functionality.

With this methodology, access is normally defined by establishing a maximum distance and people served ratios to each water point, for instance the national policies and national water sector strategy in its operational target, define access to water, in rural areas, in terms of physical coverage by establishing a maximum distance of 500m in rural and 200m to a water point and/or a proportion of urban population that has access within 30 minutes of time spent on collection of water. In the meantime, the government of Rwanda in its operation defines water supply access as one water point serving 250 people within the radius of 500m and 200m in rural and urban respectively and not queue for more than 30 minutes (MININFRA, 2016).

1.2 Problem statement

Access to safe drinking-water is essential to health, a basic human right and a component of effective policy for health protection (WHO and UNICEF, 2014). Several international policy forums have emphasized the importance of water, sanitation, and hygiene for health and development. Most recently, in 2015, countries adopted the Sustainable Development Goals, which contain a target and indicator on clean drinking water. Safe drinking water, adequate sanitation, and excellent hygiene (WASH) are recognized in the 2030 Agenda for Sustainable Development as both an end in themselves and a driver of progress on several of the SDGS, including health, nutrition, education, and gender equality.

Despite the creation and use of many indicators and approaches to assess the issue at hand, monitoring citizens' access to drinking water in Rwanda, like in other developing countries, remains a challenge. The government, through the Ministry of Infrastructure (MININFRA), and donor agencies have been creating and implementing several indicators and methodologies to evaluate water access, but figures are still unreliable, and under- and overestimations are frequently reported (MININFRA, 2021). Meanwhile, there are constantly discrepancies between official figures on water access and the actual situation from the perspective of users, which are never appropriately acknowledged. However, the recognized difficulties so far have been focused on the selection and use of water access indicators in areas such as clarity of the water access concept among and among sector-stakeholders, indicator development or selection, and how they are used to measure residents' access to water. As a result, the methodological approach to indicator selection, use, and practice on evaluating citizens' access to water as accounted for by both service providers and service consumers is examined in this study.

Another source of worry is the presence of socio-economic imbalance, both among users and spaces, as access to better drinking water sources rises with income (UNICEF and WHO, 2011). The aggregate statistics, which often characterize greater service coverage, are reported and used for planning on water-sector development programs and projects, but they fail to reflect asymmetries in citizen's modes of access to water (Cheng, 2013), obscuring the true state of water-related wellbeing in the most marginalized and low-income communities. When indicators are created at a high level of aggregation, the problem they address can be misrepresented.

Affordability is also the major limiting factor in mobilizing additional financial resources from water tariffs. However, affordability is not monitored as the global indicators and studies can be complex and unavailable. This limits the estimate of the amount of funds that could be raised through tariffs. The other key limiting factor is the users' willingness to pay, which could be lower than the capacity to pay given that users often either consider that poor service quality does not justify the price, or they have alternative cheaper options such as private wells, boreholes or open defecation.

To bridge these gaps, QGIS and excel were used to analyse the househods that are located in the boundaries of 500m from public water point as an indicator set by National Strategic for transformation (NST1) stating that every Rwandan will have access to clean water not more than 500m from their dwellings.

1.3 Objectives of the study

1.3.1 *Main objective*

The overall objective of this study is to map and analyze citizens' access to water supply services in Rwamagana district.

1.3.2 *Specific objectives*

Specifically, this study aimed:

1. To map water supply services coverage in Rwamagana district
2. To analyze mapped data and identify served and unserved area by water supply services
3. To design and conduct a willingness/ability to pay survey to determine affordability of water supply

1.4 Research Hypothesis

The following research hypotheses were formulated to guide the study.

H0: Access to improved water access is not well known due to the lack of water supply infrastructures mapping data

H1: Water supply infrastructures mapping enables to precisely measure citizens access to

improved water services.

1.5 Significance of the study

This study will benefit not only the researcher, but also UNILAK students and other higher education institutions. This study will assist the Rwamagana district, WASAC, and other WASH stakeholders.

1.5.1 *To the researcher*

The researcher gained knowledge and abilities in measuring water access in a specific community and making relevant recommendations to improve it as a result of this project. Furthermore, this research aided the researcher in his pursuit of a Master of Science in environmental information system, environmental and development studies department from Kigali University of Lay Adventists (UNILAK). This study highlighted key areas for other researchers to examine that were not covered in this study.

1.5.2 *To institutional interest*

The research will be significant to Rwamagana district officials and WASAC Ltd because of suggestions and recommendations that were provided by the researcher to help them to improve water access to Rwamagana citizens.

1.5.3 *To academic interest*

This research will also contribute to providing up to date literature for academicians and also for those who will continue with the same research, by providing knowledge from findings related to the contribution water access measurement and definition of water access key indicators.

1.6 Scope of the study

1.6.1 *Content scope*

In term of content, this research was mainly limited to analyzing citizens access to improved water supply services.

1.6.2 *Geographical scope*

With regard to the geographical context of this research, Rwamagana District, Eastern Province of Rwanda was the target area.

1.6.3 *Time scope*

To better set up the study period, the researcher recognized the nature of the study and then chose to employ 2021 datasets in terms of current water supply services and water point mapping. Data collection on assessment of improved water affordability were collected from 4 sectors from September 2021 to February 2022.

1.7 Conceptual framework

This research considered both independent and dependent variables of the conceptual framework. For the independent variables, the researcher considered ten (10) water supply schemes in rural areas and two (2) water distribution network in urban areas. The considered dependent variable was the indicators to measure water access which is related to physical water point access, its distribution systems and safety. The detailed conceptual framework was provided in Figure 1 below.

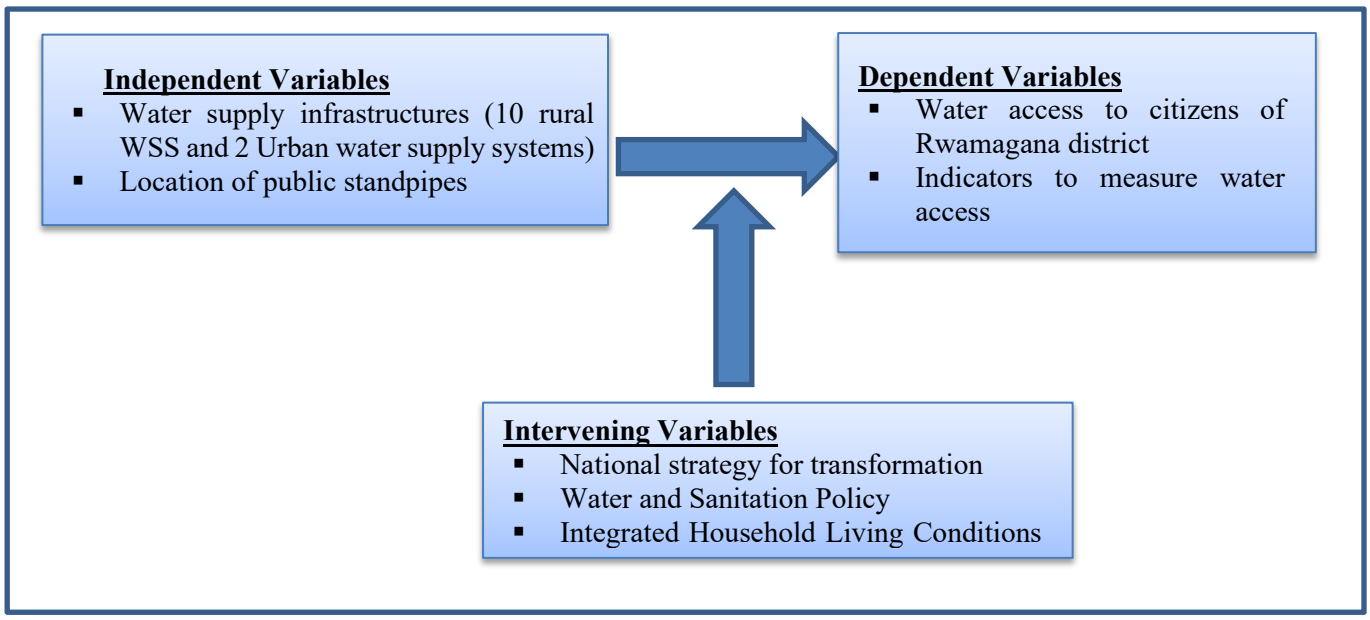

Figure I-1: Conceptual framework

1.8 Definitions of key terms

1.8.1 *Water access*

"The human right to water entitles everyone to sufficient, safe, acceptable, physically accessible, and affordable water for personal and domestic needs," according to United Nations Resolution 64/292 (Shaheed, 2014; United Nations (UN), 2010). As a result, SDG 6.1 calls for universal access to safe drinking water by 2030. The three criteria for the "Safely Managed Drinking Water" indicator are: accessible on premises, available when needed, and free of contamination (Hutton G, 2016; UNICEF and WHO, 2017). In urban settings, WHO defines safe drinking water as piped water to housing units or public standpipes within 200 meters. In rural locations, appropriate access means that obtaining water does not consume an excessive amount of time.

1.8.2 *Drinking water*

Drinking water, according to the WHO and UNICEF (2010) study, is domestic water used for both drinking and hygiene. In reality, treated surface water and untreated water from protected springs, boreholes, and wells are both considered safe drinking water. Drinking water can come from a variety of places. Water is distributed centrally by taps and standpipes, with water coming from either surface or groundwater, and is usually cleaned. At regular intervals, standpipes are installed along the pipelines. Inside a house, however, tap water is only provided at the expense of the tenant.

1.8.3 *Water governance and service provision*

Water governance relates to the range of political, social, economic, and administrative systems that control decision making with regard to water resources management and the provision of water services at different levels of a society (Jiménez Fernández de Palencia, A., & Pérez-Foguet, 2010). For instance, in Rwanda water governance is currently shaped by decentralization process partly due to the failure of centralized government service to provide reliable water service, particularly in rural areas (Masanyiwa, Z. S., Niehof, A., & Termeer, 2014). There has been a general shift from an emphasis on state water provision, public service delivery, to neoliberal ideology of market-based service delivery (McDonald, D. A., Pigeon, M., Hoedeman, O., Kishimoto, S., Azpiazu, D., & Castro, 2012), and more recently a multi-stakeholder's approach that includes community in water governance. The presence of multi-actors approach not only idealizes a symmetrical triangular interaction between the state, civil society, and market forces, but it also

pays less attention to power differences between the actors, despite the fact that power is the key player in determining who gets access to which water and at what cost (Hordijk, M., Sara, L. M., & Sutherland, 2014). Where public provision fails to reach the poor, both large- and small-scale private sellers come in, offering drinking water to the poor at costs that can be many times greater than what the government supplied well-off people (Hordijk, M., Sara, L. M., & Sutherland, 2014).

1.8.4 *Water affordability*

Affordability in water system as a global indicator compares the annual household's water expenditure with total annual income (Hutton G, 2016). However, this indicator does not take into account some key household financial recurrent costs such as for water treatment. According to Hutton (2012), affordability criterion is economically expressed by an affordability index comparing the household's monthly water cost to its disposable income or expenditure with the implication that the price paid to water services must not limit people's capacity to buy other basic goods and services guaranteed by other human rights (The UN Special Rapporteur, 2014b). In simple meaning, households must not be forced to make trade-offs between basic water and other basic needs such as food consumption or medical care costs (Langford, M., & Winkler, 2014).

1.9 Organization of the study

This research is divided into five sections. The background to the study, aims, research questions, and significance of the study were all discussed in Chapter One. The second chapter, a literature review, looked at several concepts linked to water supply and access, as well as water distribution and access in rural areas. The subject area, as well as the methodologies and/or techniques employed throughout the study, provided in the third chapter, Research instruments and procedures. The maps that identify the water distribution situation in Rwamagana district are presented in Chapter 4, Results and Discussion. The summary, conclusions, and recommendations from the current study are presented in the last chapter, Conclusion and Recommendations.

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

This chapter was designed to elaborate on relevant theories and key concepts for understanding the topic under consideration. It detailed concepts such as water supply infrastructure mapping, water access, and distribution system mapping. It came to a close with a presentation of the importance of water access as articulated in several Rwandan strategy policies.

2.1 Water supply and distribution strategic framework in Rwanda

Recognizing the key role of water and sanitation in protection of public health, socio-economic development and gender empowerment, the Government of Rwanda has committed itself to reaching very ambitious targets in water supply and sanitation, with the vision to attain 100 per cent service coverage by 2017/18, as per the Economic Development and Poverty Reduction Strategy (EDPRS) 2.

The importance of adequate water supply and sanitation services as drivers for social and economic development, poverty reduction and public health is also fully acknowledged in Rwanda's flagship policy documents and national goals. To achieve the overall Water Supply sub-sector goals, the Government of Rwanda adopted the National Policy and Strategy for Water Supply and Sanitation Services in 2010. The revised Policy presents the sector's approach on how to achieve the Vision 2020, Sustainable Development Goals (SDGs) and EDPRS objectives and breaks them down into concrete principles, objectives and statements through effective coordination among all the key players (MININFRA, 2016).

2.1.1 *Principles guiding the formulation of water and sanitation services*

According to MINIFRA (2016) the formulation of every water and sanitation services must be guided by principles of priority to basic services where each person and community has equal right to access basic water supply and adequate sanitation services. It also has to be characterized by decentralization whereby the responsibility for service delivery is vested at the decentralized level. It also builds upon community participation whereby beneficiaries of water supply and sanitation services shall be actively involved throughout the project. This can be testified by operation and maintenance costs of water supply and sanitation infrastructure which shall be borne by the users, in order to ensure sustainable service delivery. In order to promote delegated management, private

public partnerships where involving private operators is seen as the key strategy to enhance the sustainability of rural water infrastructure.

Any water supply projects shall systematically consider sanitation implications and hygiene education requirements, with activities designed and implemented in a way to ensure equal participation and representation of men and women, and to pay due attention to the viewpoints, needs and priorities of women.

The principles further state that water and sanitation sector consider service delivery in grouped settlements, taking into account the planned changing habitat structure in the entire country. It is also stated that water use should be rational and sustainable and shall abide with environmental regulations and safeguards. It emphasizes that particular attention and priority should be given to improving water supply and sanitation services to disabled people and involving them to participate in decisions concerning their interests.

Beneficiaries' participation through *Umuganda* and *VUP* activities, private operators get opportunity to sensitize and assure them to continue to deliver services efficiently and in affordable manner, and the WASAC itself to ensure it is able to meet the ever-growing demand in a sustainable and standards required. This is expected to strengthen ownership of water infrastructure by the communities (MININFRA, 2016). To promote ownership at all levels of government, the process of developing the new strategic plan to guide achievement of the development objective, the process involves the districts, the development partners working on water and sanitation sector, NGOs, private sector, Ministries, and institutions. In brief all key and non-key stakeholders have to be engaged sufficiently to contribute to the entire process.

As a result, the water and sanitation projects to be implemented through the district budget have to be identified, costed and included in strategic plan. Monitoring and evaluation systems has to be developed in conjunction with planning and budgeting procedures, involving decentralized actors (in particular the districts), in order to ensure that the activities and investments are in line with the defined sector objectives and priorities. Rwanda is politically committed to achieving long-term aspirations and targets in sustainable socio-economic development. The related targets and principles are defined in the national development flagship documents (MININFRA, 2016). The Vision 2020 aspiration states that all Rwandans will have access to safe drinking water in 2020. Improved water supply service delivery is directly related to the 'water supply and

sanitation play a critical role in preventive health care and socio-economic development in rural areas. Access to safe water and sanitation is identified as one of the 'pull factors and social benefits of living in a formal settlement. This will be accomplished through aiding districts with project planning, design, financing, and implementation. Along with greater public investment, the opportunities for community involvement and private sector management in distribution will be further explored. The goal is to have every family in Rwanda's rural districts within 500 meters of a better water source.

Appendix Table 1: Extracted from EDPRS 2 Monitoring matrix

EDPRS outcome	Indicators	Baseline	Target (2017-2024)
Increased access to basic infrastructure at urban level	Urban households within 200m of an improved water source	86.4%	100%
Increased access to basic infrastructure for rural households	Rural households within 500m of an improved water source	72.4%	100%

Source: MININFRA, 2016

It should be mentioned, however, that the 2016 baseline values (based on the results of the Integrated Households Living Conditions Survey (EICV 5)) did not account for distance. The most current EICV 5 and RDHS (2016) results reveal that when the agreed-upon distance parameters are used, actual coverage is significantly reduced. According to the most recent RDHS, 55% of rural households take 30 minutes or more to access drinking water, whereas 19% of urban households take 30 minutes or longer to receive drinking water (MININFRA, 2016).

2.1.2 Water Resource Policy and its Management Strategic Plan in Rwanda

The main drivers of water demand in Rwanda are rapid population increase, poverty, and climate change, according to the Water Resources Management Sub-Sector Strategic Plan (2011–2015). Water contamination is rampant in wetlands due to unregulated poor settlements, and storm water protection measures and disaster management are non-existent, especially when it comes to floods. Since the separation of water supply and sanitation (under MININFRA) and water resources management (under MINIRENA, formed in 2011), the separate missions are clearly defined and anchored in applicable enabling policies and plans for each water sub-sector.

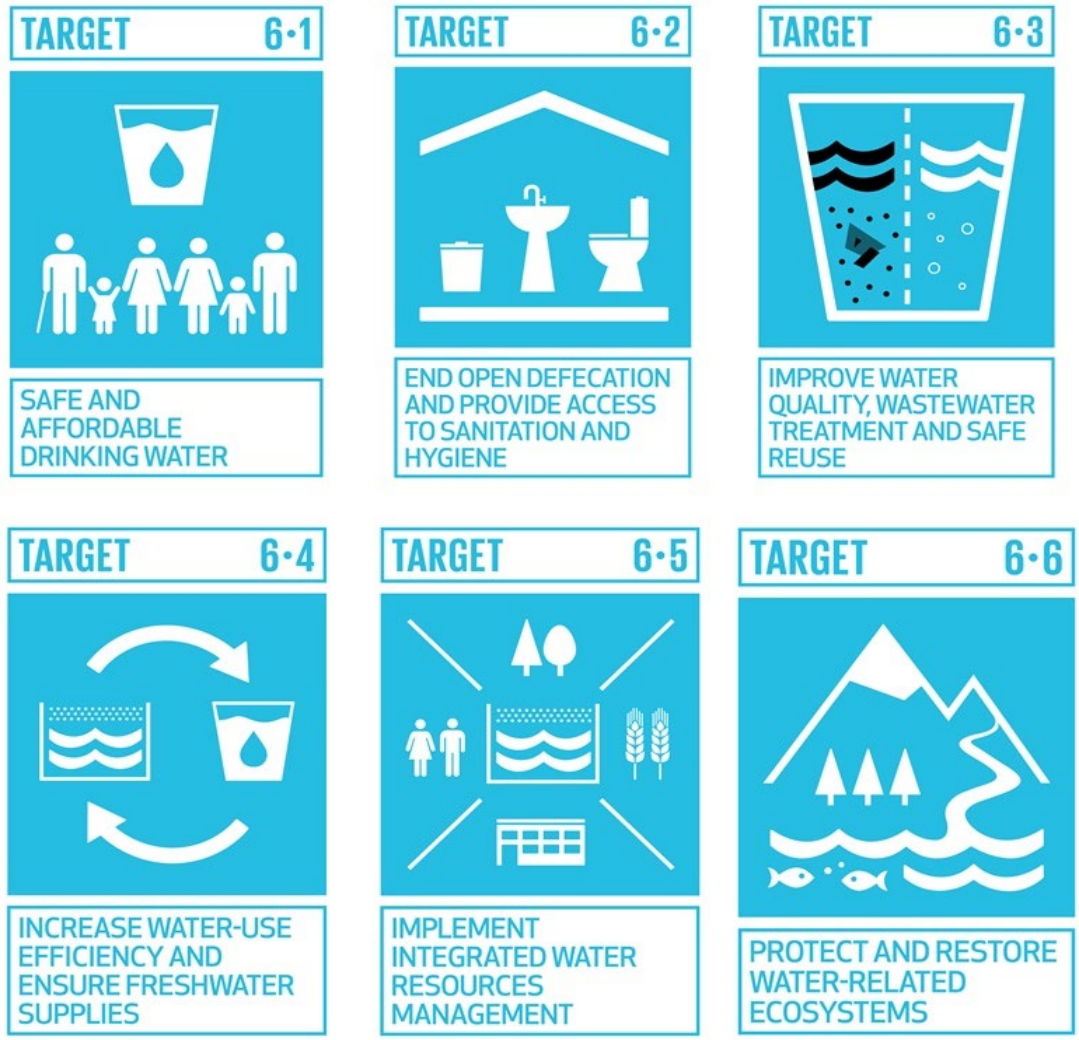

Figure II-1: Key Targets for SDG 6

Source : https://www.globalgoals.org/

2.2 Status of Rwanda's water supply sub-sector

The long-term Vision 2020 of the Government of Rwanda has been converted into a series of medium-term strategic plans, where the first phase of the Economic Development and Poverty Reduction Strategy (EDPRS I) covered the period 2008–2012 and marked a distinct change in the approach to development. Accelerating economic growth, creating new jobs, and increasing exports were prioritized. During that time, remarkable socio-economic progress was made. Rwandans have benefited from rapid economic growth, reduced poverty, more equality, and increased access to services delivery. Consequently, Rwanda has continued to achieve remarkable gains across service

delivery sectors such as health, education, business environment, energy, mining, and agriculture. The Strategic Capacity-Building Initiative that started in 2010 has further increased Rwanda's ability to translate intended priority policies into reliable day-to- day implementation (MINIFRA, 2016).

Other key achievements include the review and update of the existing Water Supply and Sanitation Policy and Strategy (MININFRA, 2016); data reconciliation within the sector in order to harmonize the concept and definition of water supply; and the establishment of a SWAp Secretariat whose structure has been agreed on by all the partners, with the next step being its operationalization. The Secretariat has already started playing a bigger role in the sector coordination. The EICV 5 survey (2015/16) indicates that 85 per cent of Rwanda's population use an improved drinking-water source. According to the most recent RDHS (2015/16), 91 per cent of the urban households and 69 per cent of rural households use an improved drinking-water source. Out of the 31 per cent of rural households that had no access to an improved drinking- water source, 16 per cent collect their water from unprotected springs, 13 per cent collect it from surface water and 2 per cent retrieve it from an unprotected spring. Regarding the time spent round-trip to obtain drinking water, 55 per cent of households in rural areas take 30 minutes or longer to obtain drinking water (MININFRA, 2016).

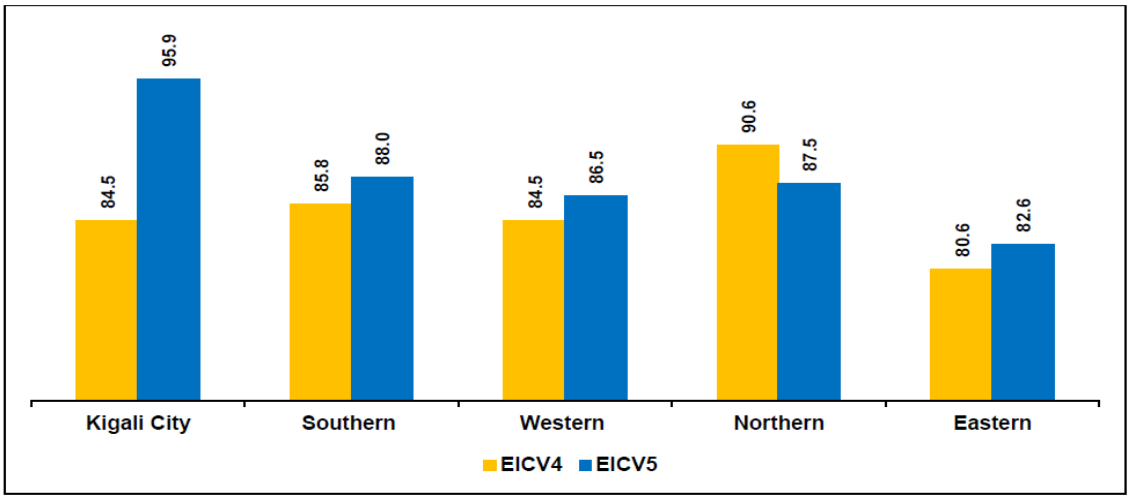

Figure II-2: Change in households with improved drinking water source (Percentage)

Source: EICV4 & EICV5. Base population: all households (NISR, 2017)

2.3 Water access as a human right

Access to water is a human right, as recognized by the UN General Assembly, with the fact that it is in itself essential for life and dignity as well as the foundation for achieving a wealth of other human rights including right to: life, food, education, adequate standard of living, and more decisively right to health (The UN Special Rapporteur, 2014a). According to the Handbook of the UN Special Report on the rights to water and sanitation, right to water is defined as "the right of everyone to sufficient, safe, acceptable, affordable and physically accessible for personal and domestic use".

It contented further that the international human rights law obliges the UN member States to work towards achieving a universal access to water guided by human rights principles and their defined standards and dimensions indicators including availability, sufficient, quality and safety, physical accessibility, acceptability, dignity, and privacy (The UN Special Rapporteur, 2014a).

However, human rights to water access like other human rights approaches to development require targets and indicators that are both quantitative and qualitative, since many essential components underpinned on goals and targets of human rights cannot be reliably quantified (Fukuda-Parr et al., 2014). These global developments have led to transformation in the governance of domestic water supply in many countries including Rwanda (Barifashe, 2014)

2.4 Review on indicators and approaches to measure water access

In recent years, there has been an increasing interest in measuring citizens' access to water within and across countries in order to quantify the national as well as global status and progress of national and international (like MDGs) water-sector interventions respectively. To monitor progress of the set goals on water access, for instance the SDG goal 6 (as defined in (WHO and UNICEF, 2012), different countries have adopted a diverse number of water access dimensions and indicators with regard to the prevailing context and conditions (locality, politics, economy, social and institutional conditions, and others). Nevertheless, keeping in pace with the monitoring the progress of SDGs as well as the legal content of the human rights to water, the common dimensions and indicators used by most countries to monitor progress on citizens' water access include the following:

2.4.1 *Proximity or physical access to water point*

The proximity to water source is normally determined through distance or walking time from user's home to the source. It is the mostly used indicator to measure the level of physical access or coverage. However, with diverse context and conditions, the optimal distance varies across nations as well as between international institutions. For instance, while WHO advocates that water source has to be within 500 meters from user's home and collection time should not exceed 30 minutes (United Nations Organization, 2014). the national water policy in Rwanda set 500 meter as a maximum distance to a water point and not more than 30 minutes collection time -go, wait, collect and return- (Govenment of Rwanda, 2017). The contents of human right to water go beyond distance and time by adding the design of facilities to suit older persons, children, person with disabilities, and to be located in safe and sound areas in terms of physical security (The UN Special Rapporteur, 2014b).

2.4.2 *Water quantity*

Water quantity entails the sufficient and continuous quantity of water for personal and domestic uses including drinking, personal hygiene, washing of clothes, food preparation, and house sanitation supplied per capita per day (The UN Special Rapporteur, 2014b). A person, regardless of being in rural or urban areas, is considered to have optimum access to water in terms of the quantity when he uses between 25 and 100 litres per day (WHO, 2011). In rural areas of Rwanda, a measure of access to water, in terms of quantity, considers an optimal water access when a household is using a minimum of 25 litres of water per capita per day through water points located within 500 meters from the furthest homestead and 80 litres in urban areas (see table below). However, the actual water uses ranges from 5 litres per capita per day in acutely water scarce areas to 30 litres per capita per day in other areas (Sphere, 2018)

Indicators/dimensions	**Standards**	**Institution**
Physical accessibility	0-500 metres or within 30 minutes in rural 0-200 metres or within 30 minutes in urban	MININFRA UN (SDG 6)
	Location with physical security	UN (Human rights)

Functional WP Density	1 water point per 250 people	MININFRA
Quantity of water	>= 20 litres per day/person in rural >= 80 litres per day/person in urban	WHO MININFRA
Reliability	24 hours per day	WHO
Affordability	5% of disposable income	World Bank
	3% of disposable income	UNDP
Quality and safety	WHO standards Acceptable odour, taste and colour	WHO, UNDP
Acceptability	Locality social or cultural standards	UN (Human rights)

Appendix Table 2: List of institutional indicators and standards for optimal citizens' access to water

Source: Researcher

2.4.3 *Service availability and reliability*

Availability and reliability require that sufficient water is supplied daily to meet people's needs now and in the future. The UN handbook for water as a human right provided further that, water should be available and reliable not only at the household level, but in all places where people spent significant amount of time including health and educational institutions, workplaces, markets and other public areas (The UN Special Rapporteur, 2014a)). In Rwanda, the policy directs that water should be available in sufficient amount of at least 20 to 80 litres per person daily in rural and urban respectively (Govenment of Rwanda, 2017)although its implementation is still a challenge owing to rapid population growth in relation to existing resources capacity constraints.

2.4.4 *Quality and safety*

Water required for each personal or domestic use must be of a quality that is safe and free from

micro- organisms, chemical substances and radiological hazards that constitute a threat to a person's health (The UN Special Rapporteur, 2014b; UNDP-IUCN, 2004). According to UNDP-IUCN (2004) the WHO guidelines for drinking-water quality provide a basis for the development of national standards that, if well implemented, will ensure the safety of drinking water. And with JMP's definition water is considered to be safe only when is collected from an improved source (WHO and UNICEF, 2012). However, this is not always the case as water may become contaminated before it reaches the tap. Yet, due to diverse context and condition across countries, measures to ensure the supply of water quality are usually defined by national or/and local standards depending on the situation of the country. In Rwanda, local service providers at the point of water production are responsible to determine water quality (Govenment of Rwanda, 2017).

2.4.5 *Acceptability*

Acceptability is crucial in determining citizens' access to water since water supply facilities might not be used if they fail to meet the social or cultural standards of the people they are meant to serve. That means all water supply facilities must be socio-culturally appropriate and sensitive to gender. For instance, gender appropriateness of water sources may be about technological design of the source or other access dimensions such as insecure location of the source and others (Masanyiwa, Z. S., Niehof, A., & Termeer, 2014). Also, water must be of an acceptable odour, taste and colour to meet all personal and domestic uses (Magloire OKOU et al., 2018)

2.5 Approaches to measure water access

One of the current discussions in measuring water access, as an intangible concept, is on methodological approaches used across countries and among stakeholders. However, it is obvious that the measurement of water access as a concept depends on the explicitness and precision of its definition, thus selection and development of indicators for measuring the concept is indispensable. Nevertheless, the precision, validity and reliability of indicators adopted to measure water access is directly linked to the availability of accessible, accurate, reliable and routinely collected and updated data (Giné-Garriga, R., Palencia, A. J.-F. de, & Pérez-Foguet, 2013).

2.5.1 *Data for indicator development, selection, and use*

The availability of data is undoubtedly linked to the development and use of indicators. It is simply not possible to develop accurate and robust indicators without the core component of high-quality

datasets with appropriate scales of measurement, while new methods and analytical tools can help to mitigate and overcome some of the issues (Wong, 2006). One of the most significant issues in the water sector in Rwanda is the lack of high-quality, trustworthy data on rural water supply that is frequently gathered, shared, and updated (JICA, 2019).

However, a number of studies have been carried out and outline a number of different techniques and tools in order to address data challenges including issue of unavailability, unreliability, and of outdated data (Bostoen, K., & Chalabi, 2006; Reger, 2017; Wong, 2006). Example of outlined tools and techniques include Water Point Mapping (WPM), the Rapid Assessment of Drinking Water Quality (RADWQ), and the Water Safety Plans (WSP). In terms of data sources, the studies advocate the use of data from household survey as well as census data since reliance on improved methods and survey data generally produce a more accurate picture than counting the number of constructed facilities (Langford, M., & Winkler, 2014).

2.5.2 *Indicators development and selection*

The measurement of many abstract concepts, for instance measuring access to water, is not underpinned by theoretically sound or policy-focused frameworks (Sawicki, 2010), thus it is becoming increasingly difficult to ignore the question marks over the choice of indicators in measuring citizens' access to water. The methodological criteria behind the 2015 SDGs framework for instance, focus on simplicity, measurability, and achievability in setting goals and targets, on statistical criteria for selecting indicators. However, the criteria were criticized being insufficient and that they are applied unevenly in practice (Hordijk, M., Sara, L. M., & Sutherland, 2014; Langford, M., & Winkler, 2014).

In recognizing some criticism of the SDGs, and setting out new criteria for targets and indicators, the "Lessons Learned" report (as deduced in (Fukuda-Parr, S., Yamin, A. E., & Greenstein, 2014)) suggests that targets should among other criteria include both absolute and relative changes, be quantifiable and time-bound, be ambitious but achievable, and be set in consultation with country teams.

The study also claims that "outcome-based indicators" are preferable to "process indicators," implying that goals should be focused on outcomes rather than means. And the most useful indicators are those that can be disaggregated to expose inter-group discrepancies and possible

discrimination, as well as those that can be measured to capture vulnerability and insecurity, as well as those that have the potential to promote data generation and gathering (Leju Celestino Ladu et al., 2018).

On the move towards Post-2015 SDGs, the UN OWG on SDGs (United Nations-SDSN, 2014)proposed that a robust indicator should, among other things, be: clear and straightforward, consensus based in line with international standards, constructed from well-established data sources, disaggregated, universal, outcome-based (but only if possible), and managed by a designated organization. However, the use of indicator criteria, such as data availability and measurability, as a "veto" over goal and target selection is more likely to exclude many human rights and human development concerns (Langford's 2012 as cited in (Fukuda-Parr, S., Yamin, A. E., & Greenstein, 2014)).

2.5.3 *Scale of measurement and up-to-date spatial data*

The choice of a spatial scale appropriate to the problem is very critical and is constrained by the existing statistical data. There is always a trade-off between the amounts of data available and the use of more appropriately defined spatial units (Wong, 2006). For instance, administrative boundaries are often used as a framework for data compilation, but they may not correspond to the ideal spatial scale of measurement for the concern at hand. In order to minimize the problems of ecologically misleading outcomes, the use of small-area units and low levels of aggregation is suggested. For instance, in Malawi WaterAid practical experience three spatial units of population were used including the census Enumeration Area (EA with population of 500-2,000 people), the Traditional Authority Area (TA comprising an average of 19 EAs), and District Area (comprising of 10 TAs) (Stoupy, O., & Sugden, 2013).

2.6 Water access indicators practice

The operationalization of indicators to measure citizens' access to water is an example of indicators practice in this context. The operation of coverage indicators on the WPM and (GI) Systems platforms has an impact on citizens' access to water. The coverage indicator and WP density indicator use data from WPM system to provide useful information for planning, management and reporting on sector- development interventions across the country (Stoupy, O., & Sugden, 2013). In order to ensure, not only the policy usefulness of indicators, but also their effectiveness several

researchers have advocated the need to institutionalize indicators (Innes, 1990; Wong, 2006). According to Innes (1990, p.232), institutionalization refers to the setting up of routine procedures and practice to enhance the continuing existence of an indicator and to legitimize the method and concept of the measure. Wong (2006) stressed further that regularizing or "standardizing" the methods and concepts of measurements prevent haphazard adjustment or manipulation of data (Barifashe, 2014; Witten, K., Exeter, D., & Field, 2003).

2.6.1 *Water point mapping*

Water point mapping, as described in previous sections, is a tool for gathering accurate data on water point functionality in which the geographical locations of all water points in a given location are acquired along with their technical and demographical data. This information is collected using GPS and a relevant questionnaire located at each water point. The data is entered in GIS package and then linked with available demographic, administrative and physical data. Then the functionality coverage information for each water point is displayed through digital maps for further actions including planning purpose and as input to inform sector-decision making. Using the provided data, this methodology can be used to generate 'physical' access indicators from the lowest geographical level (Jiménez, A., & Pérez-Foguet, 2008) and be useful for monitoring the distribution and status of water points.

2.6.2 *Water points functionality*

Functionality of the service is a very broad and complex concept. With respect to water system and the scope of this study, the definition adopted describes water point functionality 'in terms of distribution and coverage' of public water points to provide reliable and adequate water supply services in the long term (Jiménez, A., & Pérez-Foguet, 2008). Other studies consider a water point to be functional if it yields good quality water or has no technical problems even if water is seasonally available, on the contrary water point is not functional if does not provide water for more than six months of the year (Kayaga, S., Fisher, J., & Franceys, 2009)

2.6.3 *Water point coverage (WPC)*

With regard to this indicator, citizens' access to water is usually defined as the ratio between the number of people served by each water point and the maximum distance travelled by users to reach it for the service. In Rwanda, this ratio means 250 persons per outlet within a distance radius of

500m from the furthest homestead (Govenment of Rwanda, 2017) and within 30 minutes of time spent on collection of water. This implies that a citizen is considered to have an optimal access to water, only if is within a distance of 500 m from the water point or spends 30 minutes of time to fetch water. In analyzing the practice of WPC indicator, different methods have been used by different authors. For example, Barifashe (2014) used proximity-distance analysis, whereas Witten et al. (2003) used network tools to evaluate community resource accessibility to users (Witten, K., Exeter, D., & Field, 2003).

2.6.4 *Functional water point density (FWPD)*

This indicator measures water access in terms of "density or distribution of only functioning water stations per population" following this procedure. It basically computes water access as the number of water points in a particular area divided by population (within that area) expressed in number of water points per 250 people (for an empirical case example see (Stoupy, O., & Sugden, 2013). Basing on this method, in Rwanda for instance, an area is considered to have optimal access to water if its density is at least 4 WP per 1,000 (Govenment of Rwanda, 2017). This approach helps to compare and reveal the difference in the distribution of water points between different levels of geographical units of measurement.

2.7 Chapter Summary and Gap analysis

Although most of the rural population were still not be using improved drinking water sources in the past, official statistics indicate that the proportion of households having access to potable water without paying in these remote areas decreased falling from 69 percent to 43.2 per cent between 2011 and 2020 (NISR, 2018). However, population growth and other land use related issues such as irrigation have outstripped the progress in a country where the hydrological network comprising of numerous lakes and rivers as well as its associated wetlands, covers more than 10 per cent of its entire surface estimated to be 26,338 Square Kilometers.

As the history of settlement and water supply distribution policies in Rwanda as from other countries of Africa like Ethiopia, South Africa and Tanzania, there have been numerous attempts to modify the traditional pattern of settlements scattered across the countryside into one of locally concentrated settlements or model villages. These attempts started in the colonial period, with the

purpose to keep control over the population, security reasons and provision of services infrastructure to the population. In fact, the review about the history of settlement policy in Rwanda, as well as also in other countries of Africa provided an inspiration of investigating the process of relocating people from scattered to grouped human settlement. However, evidence of villagization or resettlements as a success are rather absent, and thus villagization seems to be a failing concept.

Regarding water supply distribution, the reviewed literature showed that water was given due consideration in different policies of Rwanda. However, it seems that no single research had previously focused on investigating on citizens access to water supply systems in Rwamagana district considering NST1, which gives to the present study a great value as it maps water distribution system in all sectors comprising Rwamagana district and analyze the mapped data to buffer of 500m from household to public water point. Hence this study is one of its kind as it will allow identify whether access to basic amenities such as water has improved or not in the case of Rwamagana district at every sector.

CHAPTER THREE: RESEARCH METHODOLOGY

3.0 Introduction

This chapter explained the methods and instruments which were used to collect and analyze water supply services in 14 districts of Rwamagana district, in the eastern province of Rwanda. The chapter included various aspects such as research design, description of area of the study, types of data that were collected and their sources along with data collection techniques, data analysis and processing and the research timeline of the study.

3.1 Description of the Study Area

Rwamagana District is one of the Eastern Province's seven districts. It is bordered on the north by the districts of Gicumbi and Gatsibo, on the east by Kayonza, on the south by Bugesera and Ngoma, and on the west by Kicukiro and Gasabo. It is made up of 14 Sectors that have a total area of 691.9 km2. The District of Rwamagana's hydrographic network is part of the AKAGERA basin, which is a major tributary of Lake Victoria.

The district includes two lakes, Lake MUGESERA in the south and Lake MUHAZI in the north, as well as two rivers, SUMO and NYABARONGO, and some flooded marshes such as Cyaruhogo, Bigezi, Kavura, Rwarugaju, Nyabarongo, Mutukura, Ruhita, Cyimpima, Rwandenzi, and Sumo, all of which have significant rice-growing and horticultural potential.
Lake Muhazi, which is shared by Rwamagana, Kayonza, Gicumbi, Gatsibo, and Gasabo Districts, is the district's principal water source. With an area of 872 km2, Lake Muhazi regulates the upstream section of the Nyabugogo watershed. Muhazi and Mugesera Lakes have two water treatment plants that generate 278,342.4m3 of drinking water per month.

The district is rural, and is comprised of steep hills and valleys, with springs and rivers serving as the traditional water sources. According to the fifth Integrated Household Living Conditions Survey (EICV5), published by NISR (2017), 88.3% of Rwamagana district households use an improved drinking water source. Data have shown that 48.8% of households in Rwamagana district use a protected spring. Only 28.8% of households in Rwamagana district use a public standpipe, with 11.7% of households that still use unimproved drinking water.

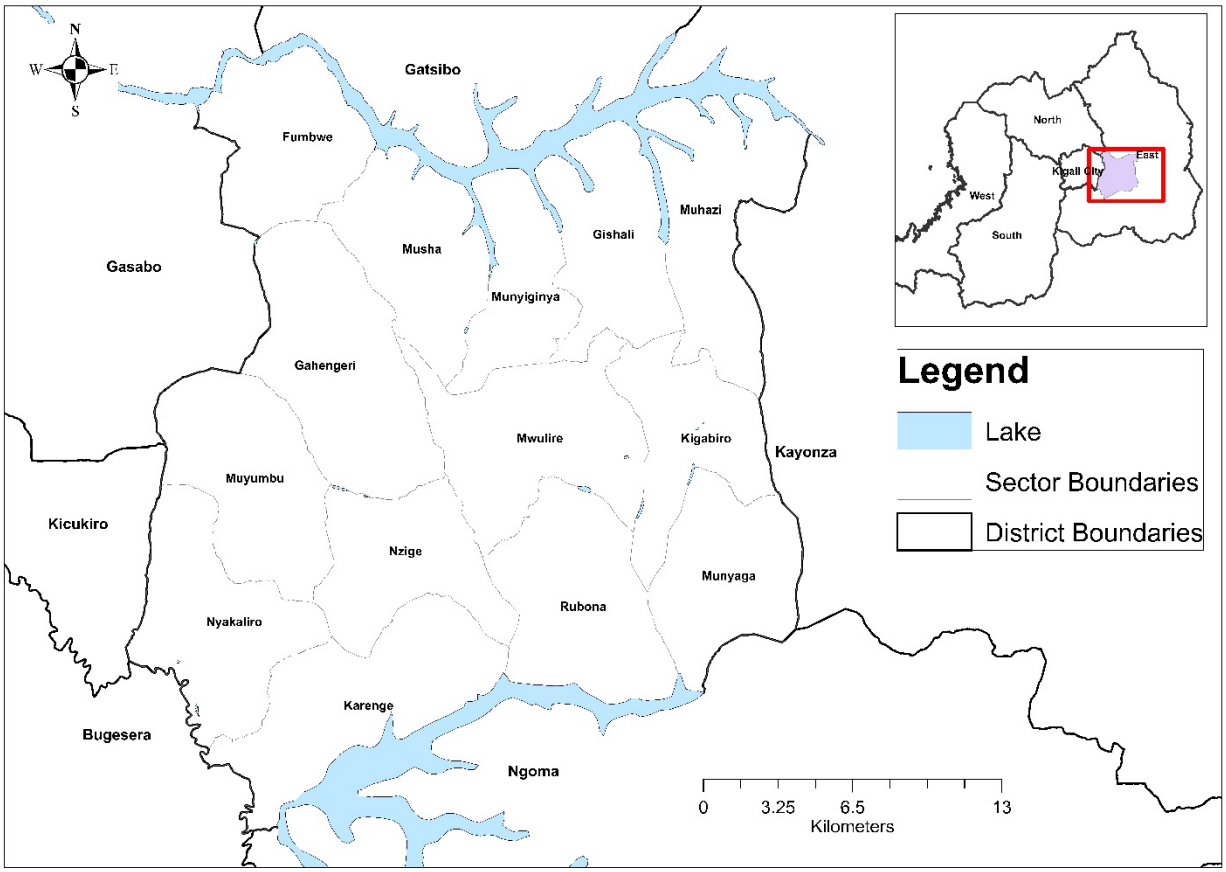

Figure III-1: Map of the study area

Source: Author, 2022

3.2 Research design

The study's major goal was to identify and map the situation related water distribution in rural and urban homes in Rwamagana district's scattered and grouped settlements. An Ex-Post Facto Research design was adapted for this, which comprises of a systematic empirical enquiry in which the scientist does not have directed control over independent variables because their manifestations have already occurred or because they are fundamentally unmanipulable. According to Salkind (2010), an ex-post facto study, also known as after-the-fact research, is a type of research in which the investigation begins after the event has occurred without the researcher's involvement.

The research was divided into various stages, the first of which was a review of the literature, which allowed the researcher to identify the gaps and create the research objectives. The tools were principally based on the use of GPS and GIS, which assisted in the mapping of the status of water infrastructure in various sectors of Rwamagana district. The adopted study design was summarized in Figure 5, which clearly detailed the essential procedures taken by the researcher to achieve the research goals.

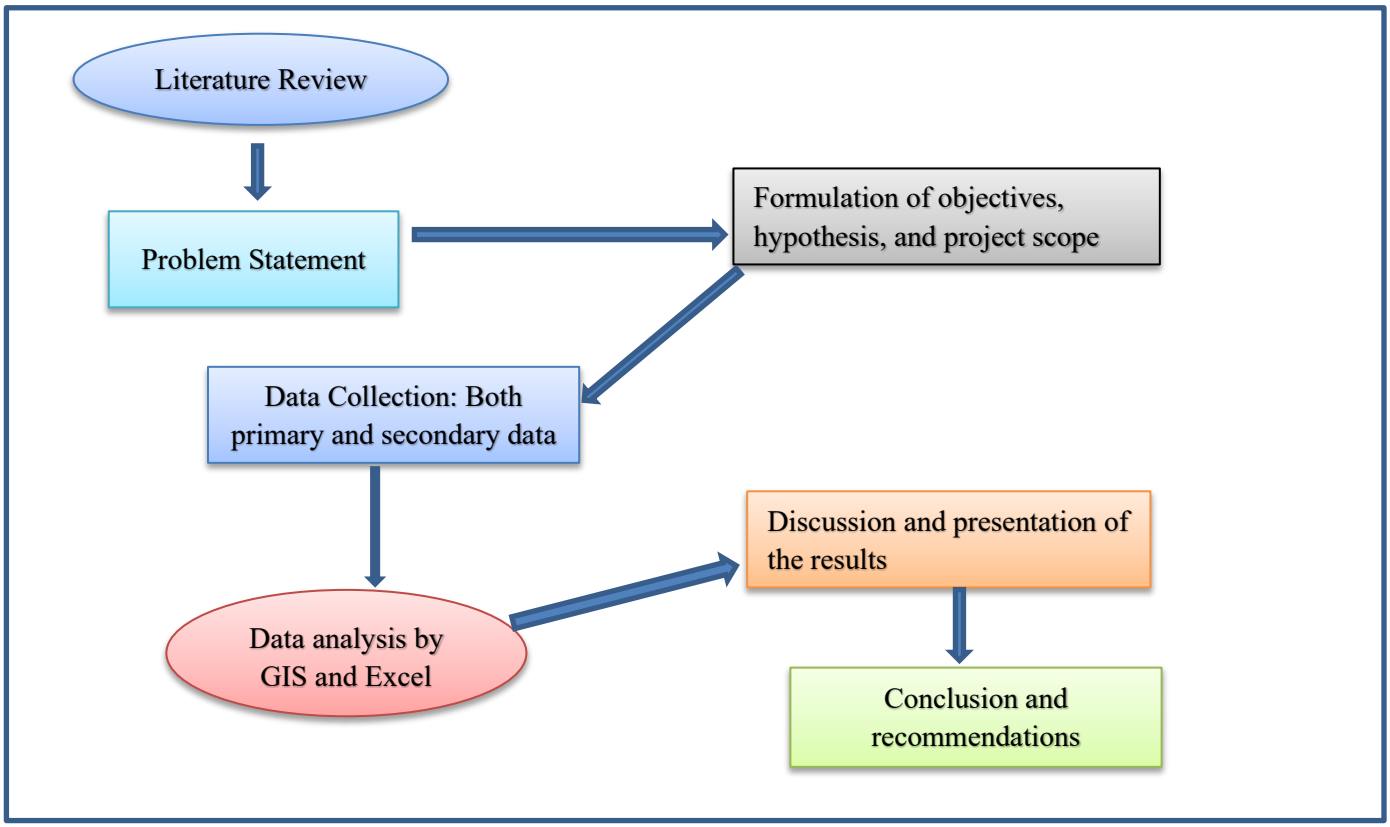

Figure III-2: Methodological framework

Source: Author, 2022

3.3 Data sources

The data for this study came from both primary and secondary sources. The researcher used a variety of data collection equipment, each of which has its own set of limitations. "Using multiple instruments of data collecting provide an effective research technique for study," Master and Kalton (1972) wrote in response to this issue. This shows that combining several data gathering techniques is a valuable research technique that may also be utilized to minimize the shortcomings of each instrument in the study.

3.3.1 *Primary data*

The field primary data were collected in two levels namely case level through household survey and mapping of water supply infrastructures in Rwamagana district. Different steps, methods and tools were employed as explained below:

- ***Household survey***

The questionnaire-based household survey was done through the prepared questions. a complete of 150 households were surveyed where by 65 percent of respondents were females. Out of 150 surveyed households, 100 households were from Rubona and Karenge sectors near Mugesera lake, 20 in Musha sector near Muhazi lake and the remaining 30 households were from Gahengeri where improved water services are not well developed as in other sectors of Rwamagana district. The household survey focused on getting information particularly on socio-demographic characteristics of the households and exploring citizens' perception and attitudes towards the practice of water access and service satisfaction.

- **Target population**

Grinnell and Williams (1990:63) defines a sample size as all people selected to take part in the research study and it should be a representative of the whole population. In this study the sample size is 150 persons as mentioned in the above section. They were all surveyed and taken to participate in the study as their number is not too big to manage.

This formula has been used to get the sample size:

$$N = \frac{z^2pq}{e^2}$$

N: sample size.
z: confidence level at 95% (standard value of 1.96).
pq: variance of population (0.501).
e: allowable error (5%).

- **Sampling techniques**

A simple random sampling technique was adopted in 4 sectors that are located near the lakes (Muhazi and Mugesera). It was chosen because we have a large enough sample size, and we know that a simple random sample has high external validity: it represents the characteristics of the larger population.

- ***Observation and mapping***

The dwellings interviewed were observed and mapped, as well as the water sources used by each household, both primary and secondary. Throughout the field data collecting period, non-participant observation, photography, and note keeping were utilized to determine the availability and use of water sources, as well as other pertinent information on the topic matter.

Both 'Q-field collection software' and handheld GPS were employed in the mapping exercise to select coordinates for the various mapped features, including the residences surveyed and coordinates for water sources specified by respondents as major and alternative sources.

According to Allan (1962), a sample size is a part of population methodologically selected for purposes of drawing a conclusion about a population and its characteristics. Spiegel (1992), observes that sampling theory is a study of the relationship existing between a population or universe and the samples drawn from it. The technique of exhaustive study (universal sampling) was used in this study as the researcher used all the targeted population to participate in the study.

3.3.2 *Secondary data*

Given the scope, time and resource constraint for primary data collection, secondary information is of great value as they can solve the mentioned limitations, be useful to supplement or cross-check primary data, and sometimes reanalysis of them may offer new interpretation although control over data quality is always questionable (Bryman, 2012). Secondary data was acquired though systematic literature review, case study documents review as well as from digital sources such as WPM (WASH-MIS) database.

- ***Systematic review***

Systematic literature review is a replicable, scientific and transparent literature review focused on a research question that tries to identify, appraise, select and synthesize all high-quality research evidence relevant to that question (Bryman, 2012). In this study systematic review was carried out to get an overview that summarize the balance of evidence on theoretical definitions and methodological questions regarding water access indicators development or selection, use and practice in determining citizens' access to water. The approach comprised the following sequence of steps.

- Define the purpose and scope of the review; the purpose was to get an overview on water

access theoretical definitions and methodological approach in water access indicator practice within developing countries.

- Develop a set of inclusion and exclusion criteria with a search strategy; only journal papers on developing countries context, Rwanda policy reports, online published materials, grey literature and other unpublished reports about water access indicators in developing countries were considered for review.
- Select studies for review; relevant studies to the scope and purpose of the review were selected.
- Analyze each study and synthesize the results; selected studies were analyzed and synthesized as presented in chapter 1 and 2.

- ***Documents review and digital sources***

Secondary data about the case study and water access matters was obtained particularly from Rwamagana district and WASAC headquarters and Rwamagana branch. From the Ministry of infrastructure report on 'national water and sanitation policy' and other documents on subject matter were obtained. Other acquired data from Rwamagana district offices included household-based survey (HBS) dataset (.csv format) of 2020/21. Administrative boundaries for eastern province of Rwanda shape files were acquired from ESRI website in digital format. Also, WPM database of 2021 was acquired from the Ministry of infrastructure through WASH MIS website. Also, the author actively participated (by asking questions) in online rural water supply network webinar workshop of December 03, 2021, which was about "Water Point Mapping in Rwanda: a reality check" by WASAC rural department and JICA through its project ''RWASOM", from this workshop the author acquired more insight on the context of WPM in Rwanda.

3.4 Data analysis

After collecting the data, data analysis process was undertaken. Text explanations and descriptions were used in case of qualitative data analysis. Data collected through field observation were analyzed in the form of pictures and explanatory notes.

In addition, primary datasets which were collected by using the Global Positioning System (GPS) were then incorporated into Geographic Information System (GIS) in order to provide maps of water distribution network in each district of Rwamagana district.

This exercise was carried out in order to identify water distribution systems in place and the related water accessibility and access situation in both rural urban sectors of district. To achieve this, the spatial analysis techniques in the QGIS software, namely the spatial multi- criteria evaluation (SMCE) techniques were used to help identify and map present water distribution with help of existing spatial data of water distribution system in Rwamagana district.

The followings described the types of data which were collected and analyzed for this research.

- ***Water Coverage***

Water points were categorized by using several definitions, but for the purposes of the constrained randomization, the researcher considered only those that were fully functional such as, improved water spring, public water taps. The nearest neighbor analysis was used to calculate the shortest geographic distance from each household to the nearest water point.

Then by using the buffer and overlay function, the households were used as the key landmark, and served for GIS layer to input vector layer and the water point GIS layer as the target point in the computations. Results were exported for analyses by QGIS software. To visualize coverage area of each water point, a buffer around each point of water point was determined to ensure water accessible area considering a radius of 500 m was produced and displayed on the maps.

- ***Water Point Mapping***

A total of 404 public water points were mapped from 12 existing water supply systems managed by both private operators (rural areas) and WASAC (urban area). The existing water point in Rwamagana district across fourteen sectors are subdivided into 45PWP in Fumbwe, 30PWP Gahengeri, 51PWP in Rubona, 50PWP in Munyaga, 21PWP in Munyiginya, 23PWP in Musha, 27PWP in Mwulire, 31PWP in Kigabiro, 26PWP in Gishali, 21PWP in Muhazi, 20PWP in Karenge, 16PWP in Nyakariro, 12PWP in Nzige and 12PWP in Muyumbu. The results reveal that public water points in rural areas are 275 which presents 68% of total public water point of Rwamagana district. The remaining 129 public water points are under management of WASAC, and they only present 32% of total public water points.

3.5 Research validity and reliability

Reliability refers to the extent to which the same answers can be obtained using the same instruments

more than one time. So, in order to ensure reliability, the GIS techniques and procedures used in producing maps were also described in chapter three of this research so as to allow other researcher to be able to reproduce the study and obtain the similar results.

Validity of research can be explained as an extent at which requirements of scientific research method have been followed during the process of generating research findings. This study was conducted in 2022 and data were gathered from September 2021 to February 2022 in Rwamagana district. QGIS was used to produce maps of water supply systems and to analyze water accessibility in 14 sectors of Rwamagana district.

The researcher also ensures the reliability of this study since the produced final document was produced according to all academic requirements and the considered case study can be based on in order to formulate relevant policies across other areas. Furthermore, the research tested the variables (independent and dependent) prescribed in Chapter one with the obtained results in order to ensure that they are in relation with the results of the study.

3.6 Ethical considerations

Participation in the household survey was voluntary and written consent was obtained from participants. The purpose of the assignment and the role of each participant was first explained to all participants, and they were asked to confirm that they understood what they have been told and expressed their free will to participate. A written briefing describing the purpose of the research and the role of participants was discussed before the beginning of the interview. The secondary data were obtained from the open sources of institutions, and everyone can access them by requesting online permission.

CHAPTER FOUR: RESULTS AND DISCUSSION

4.0 Introduction

This chapter presented the results from the analysis performed. It presented water distribution in 14 sectors of Rwamagana district and identified water access situation in each sector by locating households that are fetching water far from public standpipes more than 500m.

4.1 Water supply and service operation in Rwamagana district

It is known from the literature that amid water supply service operation and management, the central concern is whether service provision and management is mainly a government responsibility, or water is supplied and controlled by market forces (private liability), or by the community using it as a common property with open access situations (Miranda, Hordijk, and Molina, 2011). The findings in Rwamagana showed that 2 approaches were concurrently operating though in different ways and scale. For instance, the government through the ministry of infrastructures is responsible for the provision of water supply services in urban areas under its agency namely WASAC Ltd.

While on the other hand, rural water supply services were delegated to private operators for post management under private public partnership. Alongside the mentioned service suppliers, self-supply water sources were another common option found whereby citizens were collecting water from springs, although most of them were unimproved sources, and rainwater harvesting. It was clearly found that WASAC is managing big water distribution systems and water treatments plants. The main source of rural water drinking water was found that are springs and boreholes. The table below shows water management and service area:

Appendix Table 3: Water supply system management and their area of service

Name of WSS	Management (WASAC/PO)	Supplying area (Sectors)
Karenge WTP	WASAC	Karenge, Muyumbu, Nzige, Nyakaliro and part of Gahengeri
Muhazi WTP	WASAC	Kigabiro, Muhazi, Gishari
MKM	Ubuzima Bwiza MKM Ltd	Mwulire, Kigabiro and Munyaga

Fumbwe-Gahengeri	Ubuzima Bwiza MKM Ltd	Fumbwe and Gahengeri
Kirehe gravity	Ubuzima Bwiza MKM Ltd	Fumbwe
Munyiginya	Ubuzima Bwiza MKM Ltd	Munyiginya and Mwulire
Mugomero Complex	Ubuzima Bwiza MKM Ltd	Gahengeri
Byimana Gravity	Ayateke Star Ltd	Musha
Kabare-Nyabisindu	Ayateke Star Ltd	Musha
Byimana Motorized	Ayateke Star Ltd	Mwulire
Gatare-Mabare-Byinza	Ayateke Star Ltd	Rubona

4.1.1. Functionality of water connections

According to the water point mapping status digital database, Rwamagana is distributed by a total of 404 public water points (MININFRA, 2021). As can be seen in Fig. IV-1, more than one third of total number of water points were not functioning the main reasons being pump breakdown and pipes blockage, as mentioned by the water network operators during the interview. Moreover, the findings, as can be deduced from figure IV-1, showed that the number of non-functional water points managed by the private operators is higher than those managed by WASAC.

This may imply that the possible main cause for malfunctioning and breakdown of water points are likely due to lack of effective management and regular maintenance of facilities since the private operators failed to meet the maintenance cost as service were almost delivered for low cost and many citizens were reluctant to contribute for maintenance or repair cost.

However, the central reason for their reluctance may be due to the fact that they rarely use water from the public water points as will be discussed in later sections. On the other hand, most of privately operated water points were not functioning due to failure of water taps owners to settle water bills from WASAC and rural water private operators.

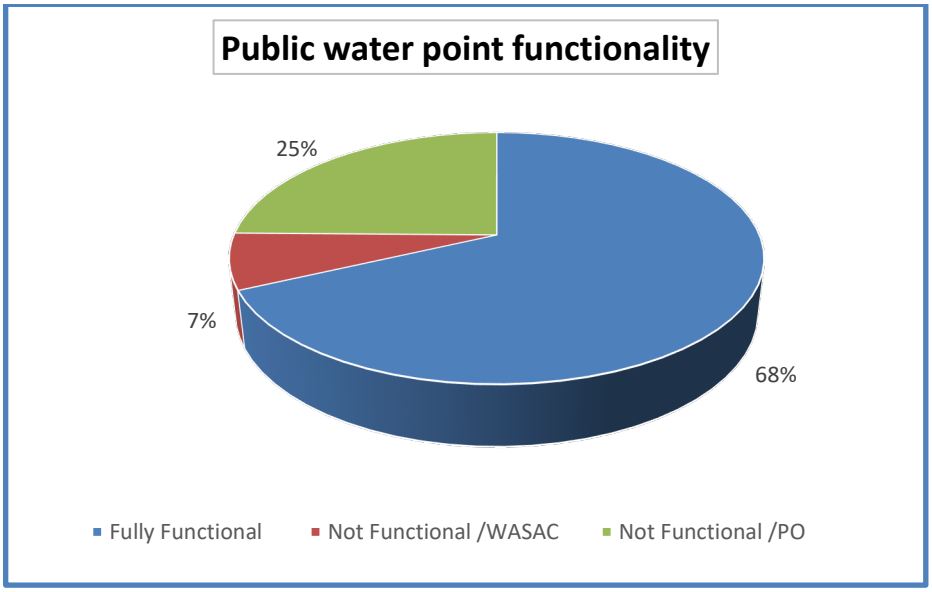

Figure IV-1: public standpipes functionality both under WASAC and PO management

(**Source:** Researcher,2022).

Figure 4.1 shows examples of functional and non-functional public water points as they were found during field data collection in Rwamagana district. However, despite that a significant number of water points were functioning, the observation made during data collection showed that a noticeable number of them, particularly publicly managed by private operators, were in deteriorated -damaged or rusted- conditions to the extent that they may supply unsafe water due to contamination. This may suggest that the availability of improved sources does not necessarily translate into supply of safe drinking water, as water may become contaminated before it reaches the users. However, according to district water and sanitation officer, the deterioration of some water supply facilities was due to salinity of ground water which was reacting with facility infrastructure as well as lack of regular maintenance down to funds limitation.

4.1.2. Households Connections to water network

Most interventions to connect households to the drinking water network are primarily concerned with physical health, but improved water access may have other important effects on household well-being by reducing the amount of time spent fetching water. Researcher conducted a detailed evaluation with support of district officials to look at the effect of private connections in a setting where most households already had access to high-quality water through public standpipes. It was found that only 17494 households (27%) are connected to piped water network. The remaining 65691 households (63%) are fetching from neighboring households or get water from public water points.

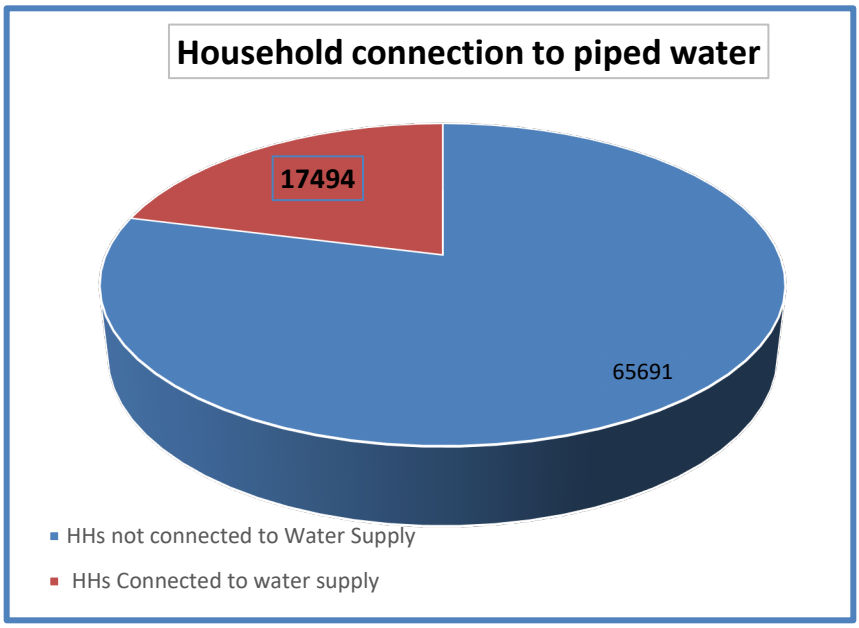

Figure IV-2: Households connection to water network in Rwamagana district

(**Source:** Researcher,2022).

4.1.3. Alternative source of water

The results from the survey and field observation showed that almost 90 percent of the households had alternative sources of water while only 10 percent located in urban areas depended entirely on main source to serve their purposes. The results show that most of households that depended solely on one main source were those who had piped water into their house compounds. It was further observed that such households used enhancement strategies by fetching water from the main source, when the supply was available, and stored them in different types of storage facilities for use at the time when the supply from the main source was not available. The reasons for using alternatives included issues such as particular purpose of water from the source for a specific use, affordability limitations, service unreliability and inconveniences, perceived low quality of water for a particular use from the main source, long distance to the main source, and other related reasons. And the most common water source types used as alternative source were firstly improved springs (72%) followed by boreholes (14%), surface water (9%) and rainwater (5%) (figure IV-3).

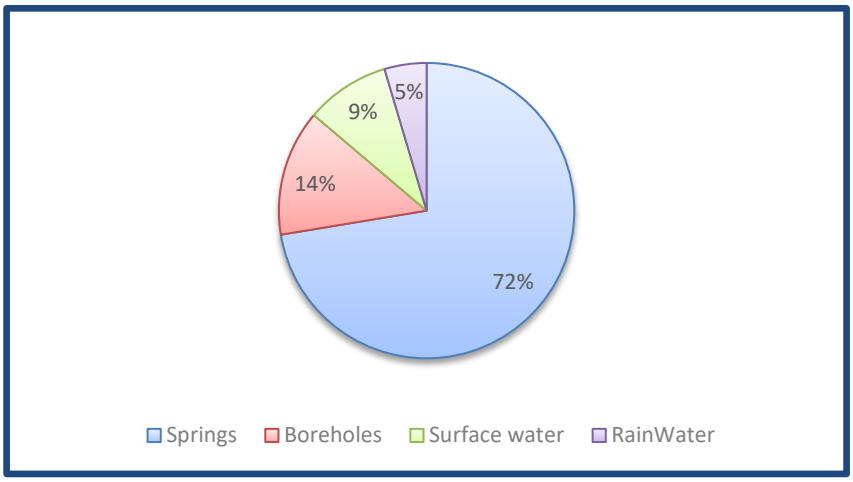

Figure IV-3: Households connection to water network in Rwamagana district

(**Source**: Researcher,2022).

4.2 Spatial distribution of public water points in each sector

As one among the aim of this study was to map the spatial distribution of water points service coverage, the functional and non-functional water points (as geo-referenced in the 2021 water point mapping dataset) was spatially visualized by the aid of GIS functions. The spatial pattern from the map showed that areas along and nearby roads and access ways as well as village centres had a better distribution of water points than areas away from the roads. This situation can be expected as public services, including water supply, are always provided where demand or users exist. And it is common that the areas close to infrastructures, such as roads, electricity, and others, to be more populated than areas away from them simply because people are always attracted to live and run their economic and other activities in such areas. The findings showed that the explained situation exists in Rwamagana district since housing units were more concentrated towards roads as well as village centers.

4.2.1 Public water point service coverage in Fumbwe sector

As outlined in the introduction that among the specific aims of this study was to map the distribution of water points and analyze the pockets of served and under-served areas, the functional water points were visualized and water access indicator that requires one outlet to serve a maximum of 250 people within a distance of 500 meter (Govenment of Rwanda, 2017) was considered. However, while considering the 2021 population data which indicates that Fumbwe sector had a total population of 28,037 (Rwamagana district, 2021) and the available 45 functioning public water

points (MININFRA, 2021), Fumbwe sector had achieved a 73 percent of full coverage.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
Fumbwe-Gahengeri and Kirehe gravity	76km	165	45	28,037	20,467

Appendix Table 4: Fumbwe Summary data

Source: Author

Yet the findings showed that a smaller number of housing units are further than 500 meters from public water points' service coverage which in turn implies that those housing units are considered as underserved. And since, housing units were used as a proxy to population data, the results may suggest that citizens found in areas out of the specified service coverage were still facing problems of access to water.

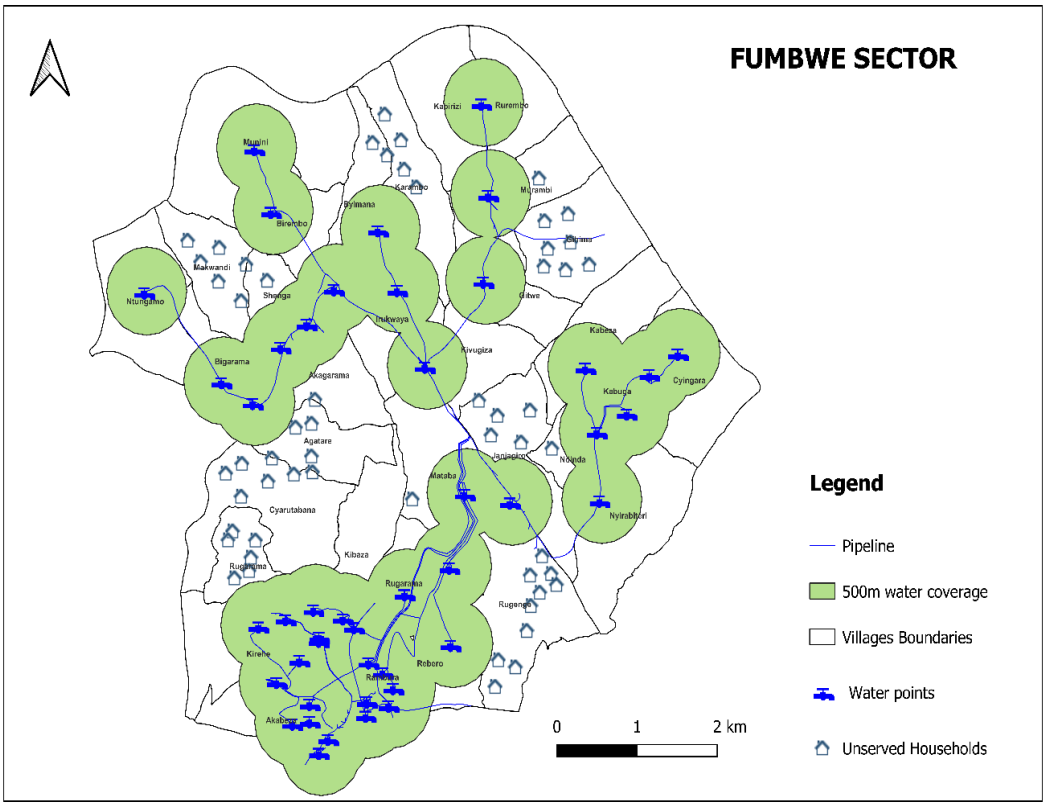

Figure IV-4: Fumbwe Water Access map

(**Source:** Researcher,2022).

4.2.2 Public water services coverage in Gahengeri sector

Gahengeri sector counted a total population of 30,384 in 2021 (Rwamagana district, 2021)and only available 30 functioning public water points (MININFRA, 2021), only 38 per cent of 47 villages are covered by improved public water point. This sector is witnessing water scarcity due to intermittent water supply which is under management of MKM Ubuzima bwiza.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
Fumbwe-Gahengeri and Mugomero complex system	42km	54	30	30,384	11,546

Appendix Table 5: Gahengeri Summary data

Source: Author

At the time of data collection, it was found that less than a half of total public water points were not functioning because of broken pipeline and bring to the whole water supply system (Fumbwe-Gahengeri) out of service. It was only MUGOMERO water supply system public water points (11) functioning.

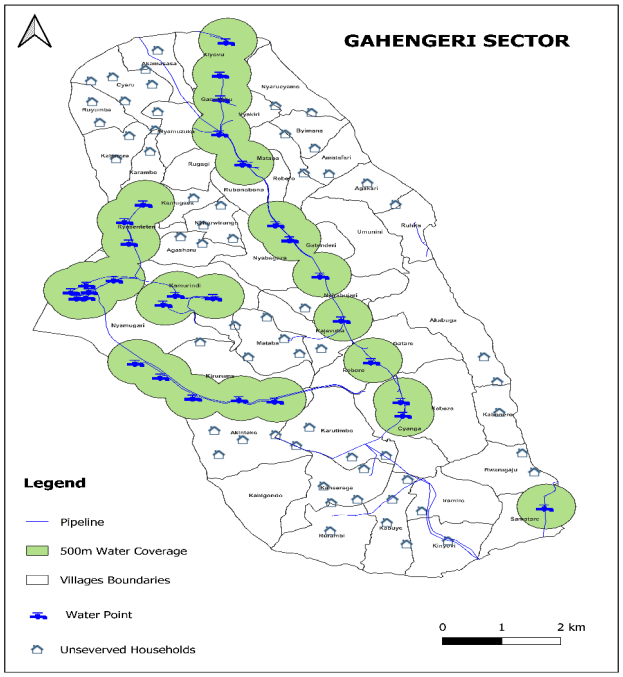

Figure IV-5: Gahengeri Water Access Map

Source: Researcher,2022).

4.2.3 Public water services coverage in Gishari sector

Gishari sector had a total population of 29,211 (Rwamagana district, 2021) and the available 26 functioning public water points (MININFRA, 2021). This sector is fully supplied by WASAC network, and many households are connected in their premises unless some people who cannot afford to pay connections fees.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
Gishari WTP under WASAC management	89km	1311	26	29,211	15,774

Appendix Table 6: Gishari summary data

Source: Author

Therefore, it was found that 54% of 26 villages that compose Gishari sector are served by public standpipes while 46% is considered as not served. It is considered at least that every village to have a public water point so that everyone can afford to use improved water.

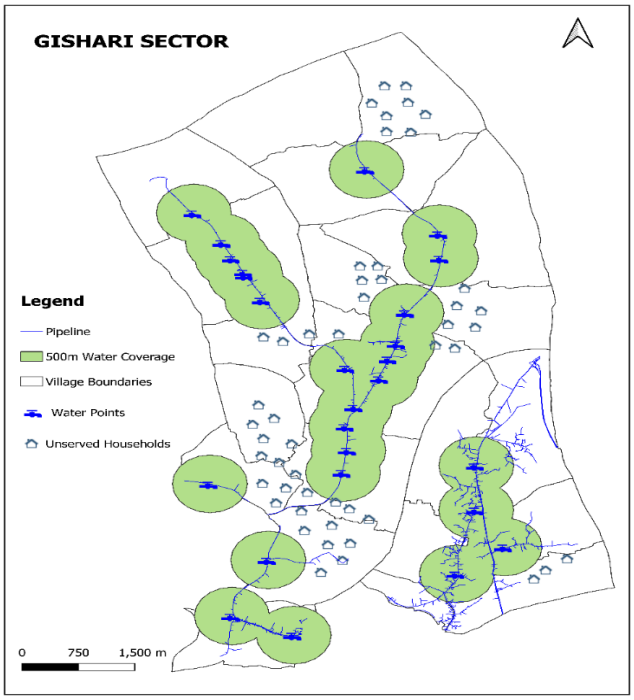

Figure IV-6: Gishari Water Access Map

Source: Researcher,2022).

4.2.4 Public water services coverage in Karenge sector

27,537 was the number of total populations of Karenge sector in 2021(Rwamagana district, 2021). they are served by 20 functioning public water points (MININFRA, 2021). The entire area of this sector is being supplied by Karenge water treatment plant under WASAC management. Many households are connected to piped water in their premises as shown in the figure below, pipes are split and distributed in different settlements.

WSS name	**Length**	**Number of HHs connection**	**Number of PWP**	**Total population**	**Served population**
Karenge WTP under WASAC management	119km	2768	20	27,537	12,117

Appendix Table 7: Karenge Summary Data

Source: Author

Only 44% of 36 villages that compose Karenge sector are covered by public water points. As this region touches Mugesera lake, people used to not pay water connection fees or pay tap manager for public water and fetch unimproved water from that lake.

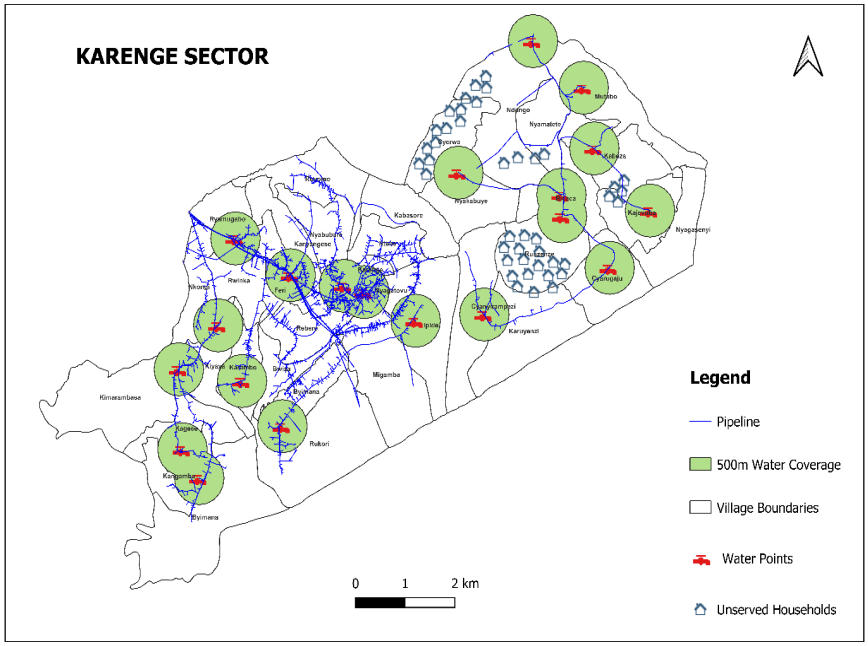

Figure IV-7: Karenge Water Access Map

(**Source:** Researcher,2022).

4.2.5 Public water services coverage in Kigabiro sector

Kigabiro sector is urban place of Rwamagana district, and it is considered as capital of the district. Demographically it had a total population of 38,865 (Rwamagana district, 2021) and the available 31 functioning public water points (MININFRA, 2021)are serving the entire area of this sector at 100 percent. Most of households have water connections in their premises unless less people who cannot afford water connection fees.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
MKM and Muhazi WTP	345km	5411	31	38,865	38,865

Appendix Table 8: Kigabiro Summary Data

Source: Author

Kigabiro sector is primarily served by Muhazi water treatment plant under WASAC management, and the small remaining part is served by MKM water supply system under management of Ubuzima Bwiza Ltd. Despite intermittent water supply due to overloaded water demand which is leading to rationing, water coverage is fully distributed in this sector.

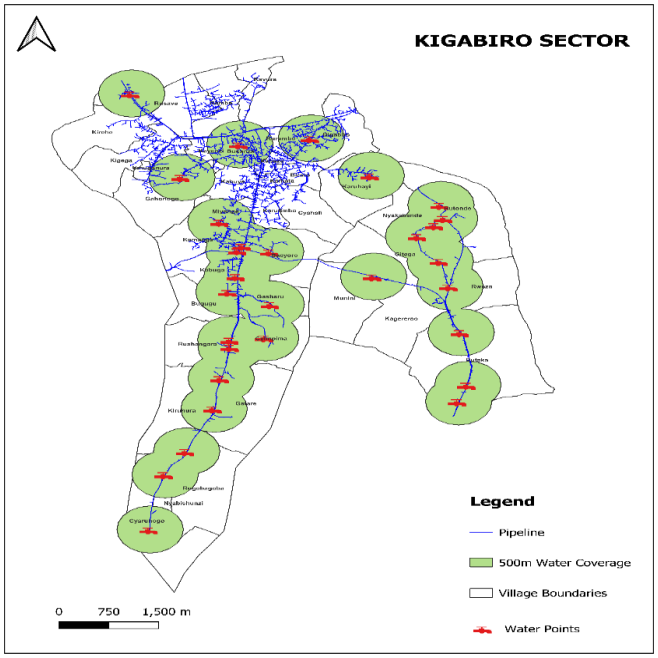

Figure IV-8: Kigabiro Water Access Map

(**Source:** Researcher,2022).

4.2.6 Public water services coverage in Muhazi sector

Demographically Muhazi sector had a total population of 31,492 in 2021 (Rwamagana district, 2021) and the available 21 functioning public water points (MININFRA, 2021), Muhazi sector achieved only 41 percent of full coverage. Among 39 villages only 16 are benefiting from public water points while the remaining 23 villages don't have public standpipes.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
MKM and Muhazi WTP	132km	1233	21	31,492	12,912

Appendix Table 9: Muhazi Summary Data

Source: Author

It is also found that people who are not connected to WASAC network in their premises are fetching water from Muhazi lake which unimproved water source.

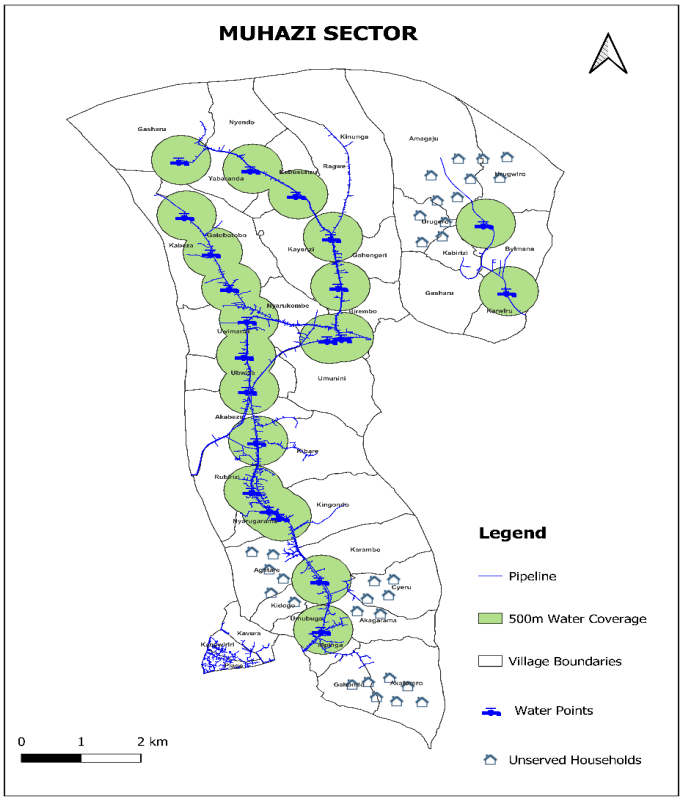

Figure IV-9: Muhazi Water Access Map

(**Source:** Researcher,2022).

4.2.7 Public water services coverage in Munyaga sector

Munyaga sector is a rural region of Rwamagana district with a total of 20728 population (Rwamagana district, 2021) and the available 50 functioning public water points (MININFRA, 2021). Munyaga sector achieved it entire water coverage at 100 per cent.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
MKM WSS	81km	167	80	20,728	20,728

Appendix Table 10: Munyaga Summary Data

Source: Author

Yet the findings showed that a significant number of housing units are all closer than 500 meters from public water points' service coverage which in turn implies that those housing units were served with MKM water supply system which fully functional unless some few maintenance issues to be performed by Ubuzima Bwiza Ltd which delegated by Rwamagana district to manage this water network.

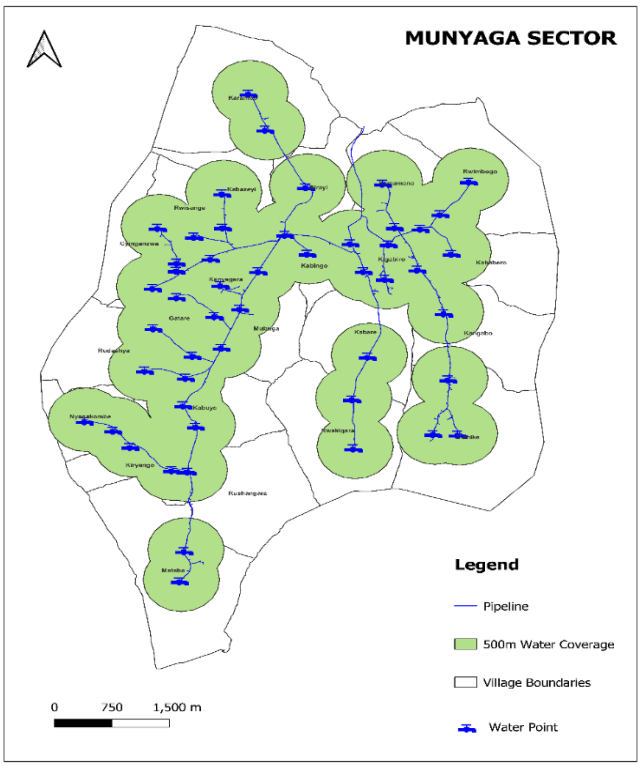

Figure IV-10: Munyaga Water Access Map

(**Source:** Researcher,2022).

4.2.8 Public water services coverage in Munyiginya sector

Munyiginya sector was statistically count of 22,836 population (Rwamagana district, 2021) and the available 21 functioning public water points (MININFRA, 2021). It achieved 50 percent of full coverage. The part supplied by WASAC network of this area is not consistently accessible due to rationing planning set to be served once a week.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
Munyiginya WSS and Muhazi WTP	34km	198	21	22,836	11,418

Appendix Table 11: Munyiginya Summary Data

Source: Author

Munyiginya WSS is under management of Ubuzima bwiza Ltd is the only water network that is supplying every day a week. From analysis, 16 villages among 32 of total villages don't have access to public water points in 500m, therefore they have an alternative water source from Muhazi lake.

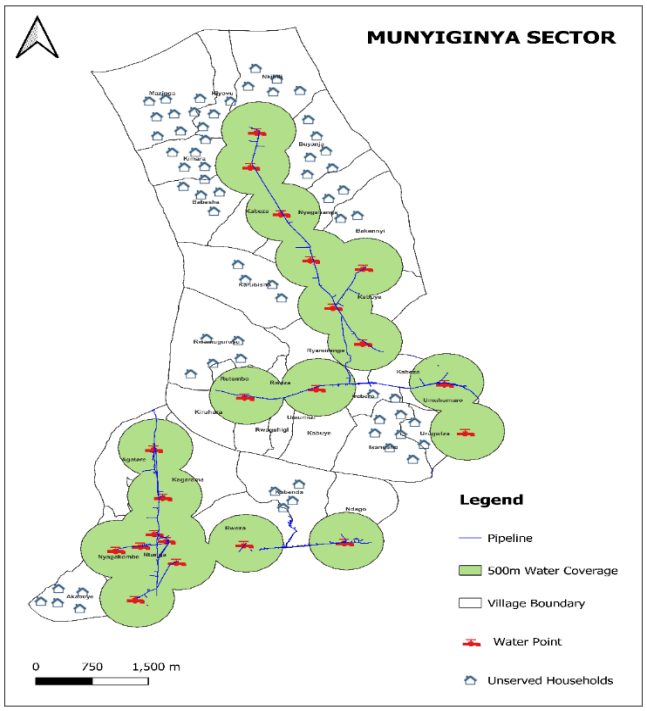

Figure IV-11: Munyiginya Water Access Map

(**Source:** Researcher,2022).

4.2.9 Public water services coverage in Musha sector

Ayateke star company limited is the private operator managing water supply systems in Musha sector. Only 23 functioning public water points(MININFRA, 2021) that are serving the total number of 26,465 populations(Rwamagana district, 2021). After GIS analysis from collected data, it was found that water service coverage is at 50 per cent but access was ranked at less than 10 per cent due to the old water supply system that is mainly serving this sector.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
KABARE-NYABISINDU WSS	43km	198	23	26,465	13,233

Appendix Table 12: Musha Summary data

Source: Author

Kabare-Nyabisindu WSS was constructed in 1986 and partially rehabilitated in 2001, it is therefore not reliable for providing water in Musha sector. The only water supply system that is covering 7 villages in 30 villages of Musha sector is called Byimana gravity.

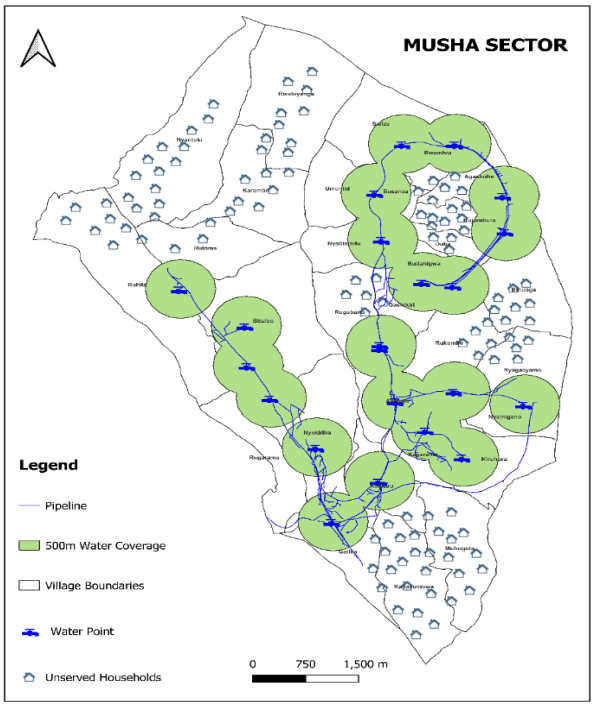

Figure IV-12: Musha Water Access Map

(**Source:** Researcher,2022).

4.2.10 Public water services in Muyumbu sector

Muyumbu sector is a peri-urban area with a total population of 32,755(Rwamagana district, 2021) but will grow to a more urban environment in the upcoming years. It is served by karenge water treatment plant under management of WASAC, unfortunately water network is unreliable and more intermittent supply due to high water demand the rationing plan is in place.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
KARENGE WTP	1231km	2106	12	32,755	9,499

Appendix Table 13: Muyumbu Summary Data

Source: Author

Muyumbu sector is being supplied by only 12 public water points and cover just only 29 per cent of the total 35 villages. After Gahengeri, Muyumbu sector is the scarce area at the time of data collection.

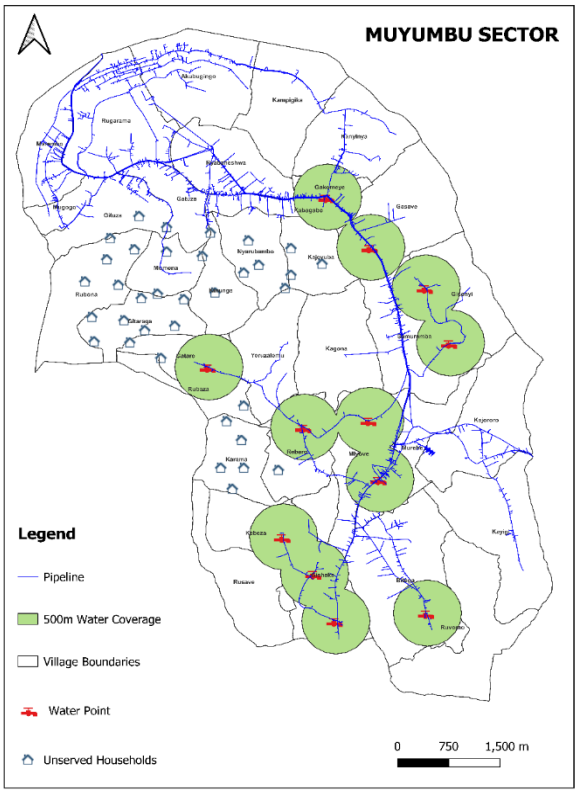

Figure IV-13: Muyumbu Water Access Map

(**Source:** Researcher,2022).

4.2.11 Public water services coverage in Mwulire sector

Mwulire sector is a special zone with many industries that require a high-water demand. It is supplied by Muhazi water treatment plant under WASAC management, and the other hand rural water is supplied by MKM WSS under Ubuzima Bwiza Ltd and Byimana Motorized managed by Ayateke Ltd.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
MKM WSS AND BYIMANA MOTORIZED	67km	87	27	28,396	14,198

Appendix Table 14: Mwulire Summary Data

Source: Author

Mwulire sector had a total population of 28,396 in 2021 (Rwamagana district, 2021) and the available 27 functioning public water points. After analysis, it was found that only 50 percent of 30 villages are served.

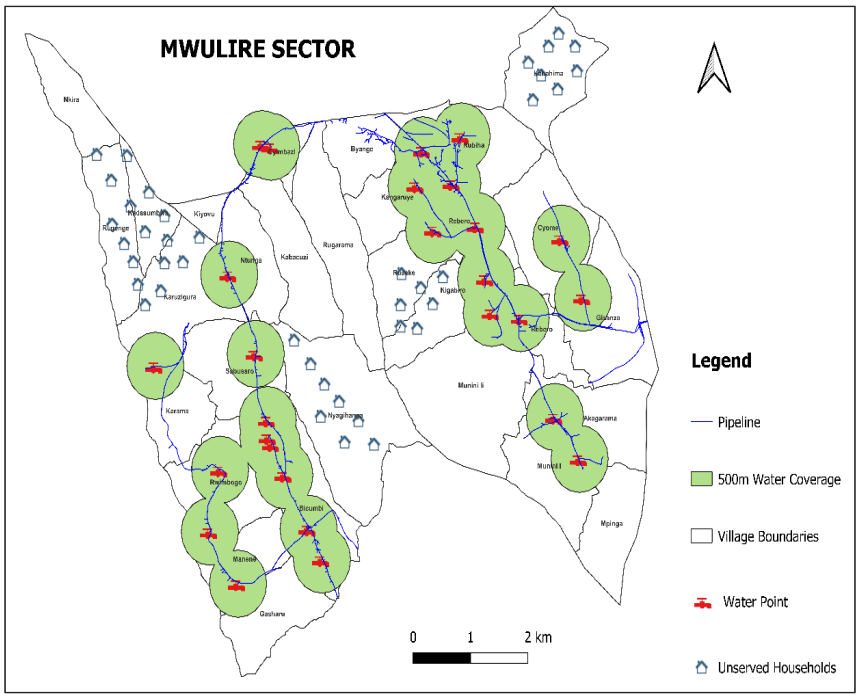

Figure IV-14: Muyumbu Water Access Map

(**Source:** Researcher,2022).

4.2.12 Public water services coverage in Nyakaliro sector

Karenge water treatment plant under WASAC management is the main source of improved water to Nyakaliro sector. Demographically Nyakaliro sector had a total population of 27,645 in 2021 (Rwamagana district, 2021) and the available 16 functioning public water points (MININFRA, 2021), It achieved only 37 percent of full coverage.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
KARENGE WTPT	452km	659	16	27,645	10,229

Appendix Table 15: Nyakaliro Summary Data

Source: Author

All 24 villages among 38 total villages of Nyakaliro sector are not accessing public water points in 500m therefore are considered as no served.

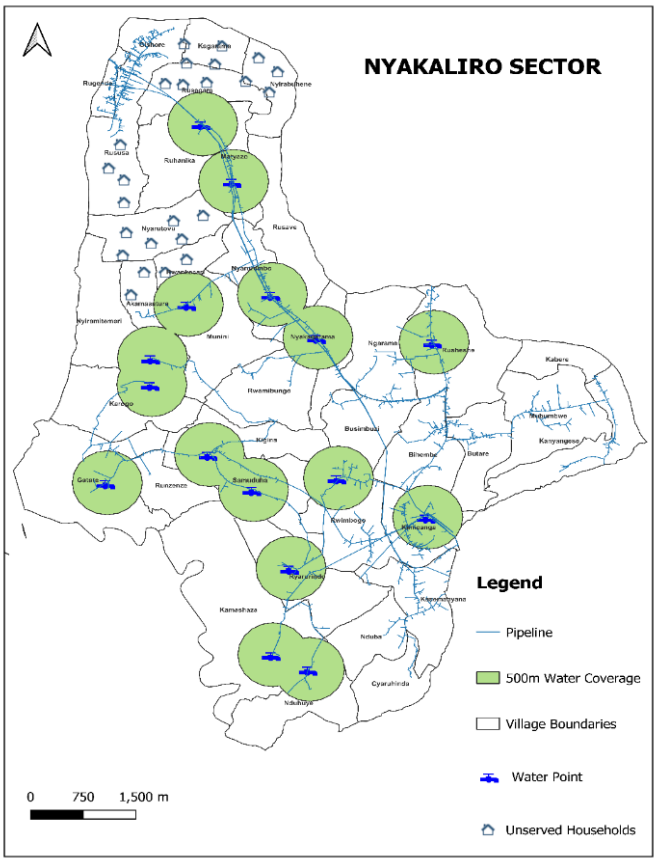

Figure IV-15: Nyakaliro Water Access Map

(**Source:** Researcher,2022).

4.2.13 Public water services in Rubona sector

Rubina sector was demographically count of 29,574 population (Rwamagana district, 2021)and the available 51 functioning public water points (MININFRA, 2021). It achieved 76 percent of full coverage.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
GATARE-BYINZA-MABARE WSS AND BYIMANA MOTORIZED	57km	53	51	29,574	22,477

Appendix Table 16: Rubona Summary Data

Source: Author

Yet the findings showed that a significant number of housing units are almost closer than 500 meters from public water points' service coverage which in turn implies that those housing units were served with Gatare-Byinza-Mabare water supply system which is fully functional unless some few maintenance issues to be performed by Ayateke star Ltd which delegated by Rwamagana district to manage this water network.

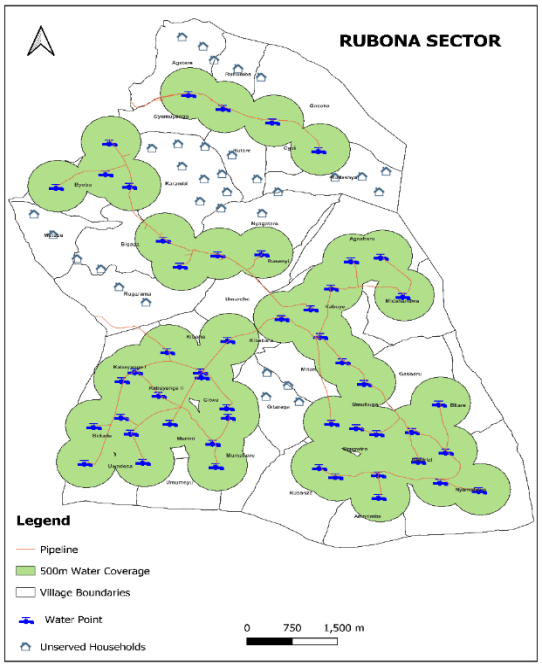

Figure IV-16: Rubona Water Access Map

(**Source:** Researcher,2022)

4.2.14 Public water services in Nzige sector

18,588 was the number of total populations of Nzige sector in 2021(Rwamagana district, 2021). they are served by 12 functioning public water points (MININFRA, 2021). The entire area of this sector is being supplied by Karenge water treatment plant under WASAC management.

WSS name	Length	Number of HHs connection	Number of PWP	Total population	Served population
KARENGE WTP	102km	654	12	18,588	7,064

Appendix Table 17: Nzige Summary Data

Source: Author

Many households are connected to piped water in their premises as shown in the figure below, pipes are split and distributed in different settlements. Only 38% of 29 villages that compose Nzige sector are covered by public water points. Water supply network is no reliable and most of population have different alternative water source mostly unimproved.

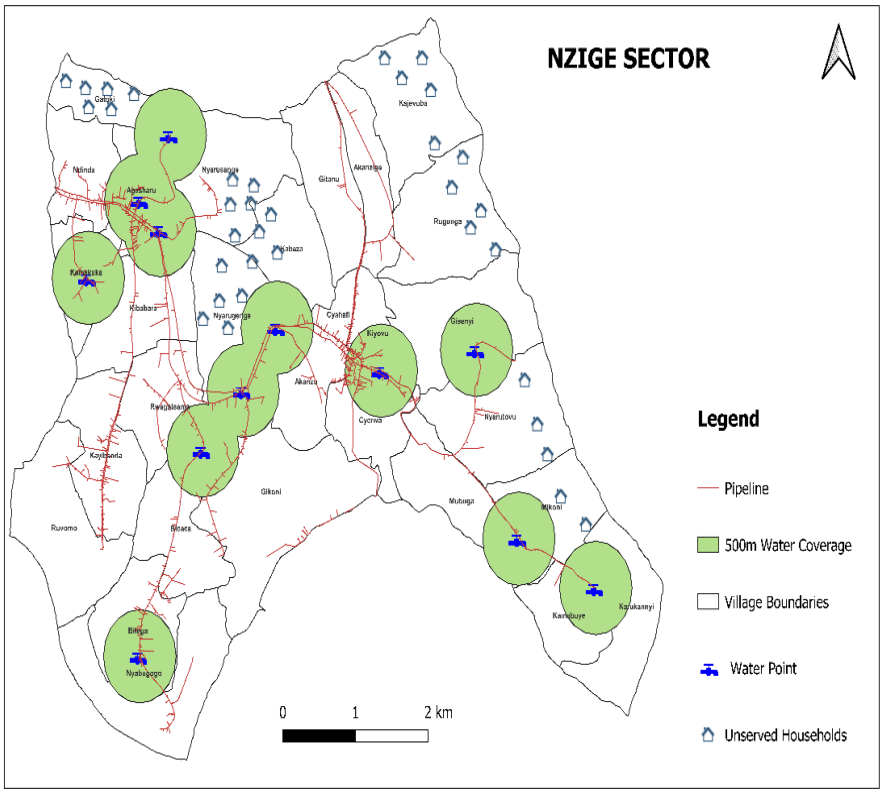

Figure IV-17: Nzige Water Access Map

(**Source:** Researcher,2022)

4.3 Affordability to water supply services

The level at which water service is affordable is often determined by the price of water in relation to the ability of the user to pay. To evaluate the level of affordability to water supply services there was a need to compare monthly average expenditure on water services and household monthly disposable income or expenditure (Barifashe, 2014; Hutton, 2012).

The results in the study area showed that 113 surveyed households were paying for the service while the remaining 37 households got water for free from boreholes and improved springs. The latter group were also divided into those who relied on public water points which offered free service or fetching water from their neighbors for free because of vendor's excuse owing to factors such as users being too old, disabled, or other social factors. The observed payment arrangement by households buying water were under two categories; first the households with WASAC piped water connections within their dwelling's compound were paying to the corporation on monthly basis 'post-paid' and the official price was explained in the tables below:

Appendix Table 18: WASAC water prices

Customer Category	Block of consumption per month	Applied tariff in Rwf (VAT exclusive)
Public tap	Flat rate per m^3	323
Residential	0-5m^3	340
	6-20m^3	720
	21-50m^3	845
	Above 50m^3	877
Non-Residential	0-50m^3	877
	Above 50m^3	895
Industries	Flat rate per m^3	736

(Source: RURA,2021)

The second category involved households buying water from delegated private operators that are managing rural water services which is higher than the prices set by RURA in urban areas. Table

below explains more about the prices that are applicable in piped water managed by MKM Ubuzima Bwiza Ltd and Ayateke Ltd depending on the type of energy used for supplying water.

Appendix Table 19: Rural Water Prices

Water system	**Tariffs VAT exclusive**		**Tariffs VAT inclusive**	
	Tariffs RWF/m^3	**Tariffs RWF/Jerican**	**Tariffs RWF/m^3**	**Tariffs RWF/Jerican**
Gravity	287	7	338	8
Electricity Pumping	731	17	863	20
Diesel pumping	921	21	1087	25
Turbo	689	16	814	19
Complex system	597	14	704	16

(Source: RURA,2021)

4.4 water supply services reliability

In this context, reliability refers to the regularity or consistency of water supply service duration. According to national water policy and other national directives objectives water have to be supplied daily, meaning 24 hours a day (Govenment of Rwanda, 2017) given the fact that water is needed all the time. However, the results in the study area showed that the supply of water from the main sources mentioned was neither "24 hours a day" nor "some hours of each day" but rather "some days of each week". They showed further with about 88 percent of the mentioned main improved sources, particularly WASAC piped water, water was supplied on average of some hours of 1 to 2 days a week.

Nevertheless, the public water points, boreholes and springs which served for only 5 percent of the surveyed households as main source, water was available 24 hours daily. And about 7 percent of households who depended on unimproved sources such as open dug well were getting water inconsistently since water in those sources were depending on seasons of the year. Further observation showed that during the days when water was available, the supply was usually during night and rarely during the daytime.

4.5 Quantity of water consumed

The national government through the Ministry of Infrastructures and other donor agencies had a target of providing at least 20 litres of water per capita per day as a minimum basic level of service in rural areas of Rwanda (Govenment of Rwanda, 2017). The findings from the study areas showed that the mean quantity of water, from improved sources, used per capita per day was about 29 litres, the amount above the minimum basic service level specified by the government with the range from 20 litres to about 80 litres per capita per day. However, the distribution of data is somewhat symmetrical as scores are skewed at 0.335 whereby the median is 29 similar to the average quantity and close to the mid-spread of 20 litres.

According to the statistical results, about 60 percent of surveyed households met and were above the national target of using at least 20 litres per capita per day while 40 percent were below to far below the set national target. However, there are several possible explanations for the category that were below the national benchmark including factors such as limited affordability, water supply unreliability and access inconveniences, and others. The results showed that about 47.5 percent of the respondents were satisfied with the quantity of water used while 37.5 percent were not satisfied.

They indicated further that majority of those who were satisfied were using the quantity of water beyond the national average and collected water from house connection, public standpipes, or neighbor's taps of within 15 minutes collection time. However, the households that were not satisfied on the quantity of water they used highlighted financial capacity as a major constraint for accessing adequate quantity of water for household's domestic uses.

4.6. Discussion

The present study was concerned with mapping and analyze citizens access to water supply systems in Rwamagana district, Eastern Province of Rwanda. It focused on the selected villages all 14 sectors of the district. The results revealed that Muyumbu, Nyakaliro, Nzige, Karenge, Muhazi and Gahengeri sectors are the most less served at 29%, 37%, 38%, 41%,44% and 47% respectively. This study was limited to evaluating the relationship between grouped settlement and improved level of access to clean water in Rwamagana district. The maps indicate water distribution systems in different sectors as indicated in the figures in chap IV. The study produced maps indicating the different types of water distribution systems in the selected study sites. It is commonly established

that water is an essential component to ensure equitable, sustainable and productive rural economies. It is required for guaranteeing food security, health and personal hygiene, as well as for agricultural production and energy supply, among others. Sustainable water management, sufficient water infrastructure and access to a safe, reliable and affordable supply of water and adequate sanitation services are key to improving rural livelihoods, expanding local economies, creating decent jobs in rural areas and across economic sectors, and maintaining a healthy and productive workforce. A failure to address issues related to water runs the risk of having negative repercussions on rural economies, livelihoods and decent work. Water is recognized as a human right that "entitles everyone to sufficient, safe, acceptable, physically accessible and affordable water for personal and domestic uses" (CESCR, 2002).

The cornerstone of any healthy population is access to safe drinking water. Water contributes much to health. Good health is the essence of development. The importance of adequate water supply and sanitation services as drivers for social and economic development, poverty reduction and public health is also fully acknowledged in Rwanda's flagship policy documents and national goals. Water contributes to health directly within households through food and nutrition, and indirectly as a means of maintaining a healthy, diverse environment. The right to water was defined as "Everyone's right to have access to sufficient, safe, acceptable and physically accessible and affordable water for personal and domestic uses". Safe drinking water is a relative term, varying with country and international organizations' specific quality standards and guidelines (Dinka, 2018).

However, access to an improved water source is more about ensuring the availability of an essential quantity of water nearby (according to the Joint Monitoring Program, availability of at least 20 L per person per day from an improved source within 500m from the user's dwelling) than to guaranteeing that the water is safe or adequate for human consumption. One cannot thus ensure that there is a direct link between an improved water source and water quality (Smiley, 2017).

The findings from of this study revealed that most of the sectors had water accessibility rate which was less than than 50% on average. This is not in line with the NST1 keys strategic interventions about the water accessibility, which stipulates that water accessibility will be scaled up from 85 % (Estimates 2017) to 100% by 2024 (GoR, 2017). It also indicates that there is still much to be done to increase water accessibility to the desired level. According to Patunru (2015), improved water sources do provide some degree of additional protection over unimproved sources, despite the fact that microbial quality cannot be guaranteed (Shaheed et al., 2014). In fact, the definition of improved

sources does not consider aspects such as physico-chemical contamination and insufficient mineral quality.

Demand for water will continue to increase, and it has been estimated that by 2030 nearly half of the population will live in areas of high-water stress, which will result in the displacement of populations. While water scarcity is likely to limit opportunities for economic growth and the creation of decent jobs in rural areas, the increased demand for water in areas with reduced water availability or high competition for water calls for an increased diversification of water sources, such as low yielding wells and springs, rainwater or storm water harvesting, urban runoff, and wastewater recycling. This not only has the potential, through technological development, to create jobs in the operation and maintenance of treatment plants to reclaim water, but it enables new forms of small-scale intensive uses of water such as the cultivation of highly profitable crops in small plots.

At the end, all the specific objectives were achieved by either analysis and presentation of the second data collected from the institutions responsible for water supply systems and analysis of surveyed data from 150 households.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.0 Introduction

This chapter presents the conclusion of the study and recommendations related to the findings of the study.

5.1 Summary of results

The present study was concerned with mapping and measuring citizens access to water supply in Rwamagana district in the eastern Province of Rwanda. There are abundant water resources which could be explored to ensure access to water among its residents. Nevertheless, the existing distribution systems in some cases, do not reach those in need because some of them are too aged and need rehabilitation. Both primary and secondary datasets were employed. The study identified different types of water supply systems under different management in 14 sectors namely Fumbwe, Gahengeri, Rubona, Munyaga, Munyiginya, Musha, Mwulire and Kigabiro, Gishali, Muhazi, Karenge, Nyakariro, Nzige and Muyumbu of the Rwamagana district.

The results were presented into maps that revealed that most people from Rwamagana district are getting water from public water points where 29.1% households members use water from public standpipe and get it within 500m and 30 min round trip while 27.3% household members use water from public standpipes but get it using more than 500m and more than 30 min round trip. Muyumbu, Nyakaliro, Nzige, Karenge, Muhazi and Gahengeri sectors are the most less served at 29%, 37%, 38%, 41%,44% and 47% respectively. It was revealed in the household survey conducted that water supply supplying those sectors are not reliable and are too overloaded and old to serve the big population in that area. Kigabiro and Munyaga sectors found as fully covered 100% and the most people are connected to their premises.

5.2 Conclusion

As discussed previously, for the government better access to water exists when a household collects water within a distance of not more than 500 meter while spending a maximum of 30 minutes for a round trip, with the consumption of at least 20 and 80 liters per person per day in rural and urban respectively. Also, as the service provider, an optimal water access to the

government is when one (1) public water point serves 250 people within 200 and 500 meters in rural and urban areas respectively. This objective definition interprets water access in terms of service “coverage” and management. On the other hand, the findings showed that among users’ better access to water is likely to mean “actual use of water for an intended purpose”. The definition goes beyond physical access to water as well as service coverage issues, as the evidence confirmed that a mere provision of water supply facilities does not necessarily indicate actual use by the intended users. However, in the national context, the findings can be useful in the application of remedy policies on water sector by providing a better understanding on "who counts what and how" when measuring citizens' access to water. This is particularly useful for sector practitioners and policy makers as it provides inputs necessary for monitoring the progress of national and global goals as well as water-sector development.

5.3 Recommendations

The study recommends the following to:

5.3.1 *To Policy makers*

RURA must Conduct regular and ad-hoc inspections to ensure compliance with license obligations access to basic water supply services and ensure that service is well provided for meeting sustainable development goal 6. It is also recommended to amend the regulations and ensure they contain measures to enforce non-compliance with license terms and obligation.

MININFRA is recommended to regular monitor the implementation of national strategic for transformation and national water supply policy stating that by 2024, every Rwandan will have access to clean water in 500m, 200m in rural and urban areas respectively and 30min round trip.

5.3.2 *water service providers*

In the effort to implement national water policy, water services providers should Procure adequate store of spare parts for routine maintenance of managed water supply systems for sustaining the existing infrastructures. They are also recommended to achieve the annual targets of extending water network in order to reach the big population and connect more households to their premises.

5.3.3 *To other researchers*

- Research that can investigate on the accessibility and on challenges encountered by water consumers living in urban and rural areas can bring new insight to this research.
- There is a need to complement this empirical evidence regarding the quality of water accessed and on frequency of water availability, using field survey.
- The study further recommends further research on water accessibility, and that it should focus on issues such as water availability and quality of water in different settings.

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APPENDIX

Consent form

Greeting and reading of rights: this statement is to be read to an identified respondent taking part in the survey before the interview.

My name is ……, I would be grateful if you could spare a little of your valuable time to complete this questionnaire. This questionnaire was prepared and designed to collect information related to the means, practices, and strategies currently in use measure water supply access. The information provided from this survey will serve as a baseline and will inform the decision to plan for some activities to increase water access in Rwamagana district.

Taking part in this assessment is totally your choice. You can decide not to participate, you are even allowed to stop taking part in this survey at any time for any identified reason.

Before we start asking you any question, we will request you to give us your verbal consent. Rest assured that any information you provide will be kept strictly confidential to research thesis.

During the survey, I may take some pictures if you allow it.

You can ask me any question that you have about this survey before you decide to participate or not. In case, you do not understand the information or if your questions were not answered at your satisfaction, please do not declare your consent on this form. Thank you!

Name of Respondent…….....

Position of respondent in a family…….

Date of interview……

Signature……

Questionnaire of household interview

General Information

1. Sector name…… Cell name…….
   Village name……

2. Name of interviewer…….

3. GPS coordinates (house location): ……...

   a. X-coordinates: ……...

   b. Y-coordinates: ……...

Household details

4. What is your position in the household?

   a. Husband [] b. Wife [] c. Other (specify)…….

5. What is the number of people in your household? ……..

6. How many bedrooms are in this house? …….

7. Is the house owned by the household? Yes [] No []

8. Does your household own the following assets in working condition? (Circle the answer)

   a. Mobile phone b. Television c. Motor vehicle c. Motorcycle d. Bicycle e. Others (specify please) ……

Water service provision and access

- ***Water sources, proximity, and costs***

9. What is the main source of water used by the household? (Circle the answer)

   a. Piped water into house/plot/yard

   b. public water point (hand pump/tap/kiosk)

   c. Neighbour's tap/standpipe

d. Protected dug well

e. Open dug well

f. Surface water (e.g., river, dam, lake)

g. Others (specify)…….….

10. Where is the main source of water you are using? (Specify the name of the place or facility)? ……….. ……

11. What are the uses of water collected from the mentioned source? (Multiple answers are possible). a. Drinking b. cooking c. Dish washing d. Bathing e. laundry f. Others (please specify…….)

12. How long does it take to reach the source, get water, and come back? (Walking time in minutes):
……

13. How do you consider the time spent for fetching water?

    Very short [] Short [] Moderate [] Long [] Very long []

14. Who is the service provider of the water you are using?

    a. WASAC []

    b. MKM Ubuzima Bwiza []

    c. Ayateke Star Ltd []

    d. Others (specify).....

15. Who is responsible for collecting water from the source?

    a. Husband [] b. Wife [] c. Female children [] d. Male children [] e. Others (specify please) ……

16. Do you pay for water from the sources you mentioned in questions 12 above? Yes [] No []. (If yes, ask question 18 and 19 and if no, jump to qn.20)

17. How much money (in Frw) in average do you pay for water supplied per household per day for each source?

a. Main source Frw……...

b. Alternative source(s)…….…….

18. How do you consider the cost of water supplied?

Very low [] Low [] Moderate [] High [] Very high []

19. Is the main water source the nearest source from this house? Yes [] No []. (If no, ask question 21).

20. Why not using the nearest water source? (Multiple answers are possible)

o High water price []

o Low discharge []

o short duration of supply []

o Inappropriate time of supply []

o User long queue []

o Other (specify): …….

Water supply quantity and reliability

21. What is the average quantity (in 20litres bucket) of water does the household use per day?.....….....

22. How satisfied are you with the quantity of water you are using?

o Very satisfied []

o Satisfied []

o Neither satisfied nor dissatisfied []

o Dissatisfied []

o Very dissatisfied []

23. If not satisfied, what quantity of water will you consider enough?.....

24. How many hours per day water is supplied/available……?

25. How many days per week water is supplied?.....

26. At what time of the day water is supplied? Day time [] Night-time [] Both []

27. How satisfied are you with the frequency of water supply or availability?

    o Very satisfied

    o Satisfied

    o Neither satisfied nor dissatisfied

    o Dissatisfied

    o Very dissatisfied

28. Is the water availability varying with months? Yes [] No [] (If yes ask question 30 if No go to qn. 31)

29. Which months are marked by scarcity of water supply? (If any selected ask qns. 29)

    J F M A M J J A S O N D

30. How long does it take to get water from that source during water-scarcity period? (Walking time- go, wait, and return in minutes):
    …….....

31. In general, how do you consider your access to water supply services?

    o Very satisfied

    o Satisfied

    o Neither satisfied nor dissatisfied

    o Dissatisfied

    o Very dissatisfied

32. From your opinion, what aspects do you consider important for an optimal access to water……

Thanks for your co-operation!